Libertinus and diacritics

Anchor model

Libertinus has an anchor model where the base anchors are set to a location where the mark would be placed relative to the base. See the blue dots in the left panel of Figure 1; from top to bottom, the above-anchor is the blue dot above the base glyph, the overlay-anchor is blue dot inside the base glyph outline, and the below-anchor is the blue dot below the base glyph. The mark anchor is generally near the optical center of the mark glyph outline. See the red square in the middle panel of Figure 1; the mark's below-anchor is at the optical center of the mark. The base and mark are combined in the right panel of Figure 1 by superimposing their below-anchors.

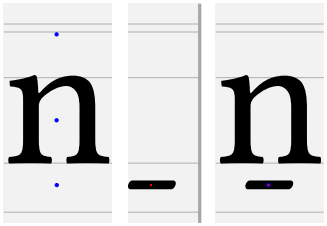

Figure 1: Libertinus n and macron with anchors.

Other fonts use a different geometric model for anchors. In Noto, the base's below-anchor is at the baseline, the overlay-anchor is inside the glyph, and the above-anchor is at the x-height. See the blue dots in the left panel of Figure 2; from the bottom to the top, the mark's below-anchor is the blue dot on the baseline, the overlay-anchor is blue dot inside the base glyph outline, and the above-anchor is the blue dot at the x-height. In Noto, the mark's below-anchor is at the origin $(0, 0)$, and the mark glyph outline is drawn below the origin. In the middle panel of Figure 2, the mark's below-anchor is the red square at the origin. The base and mark are combined in the right panel of Figure 2 by superimposing their below-anchors.

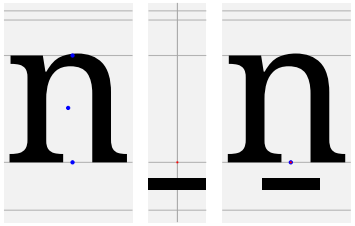

Figure 2: Noto n and macron with anchors.

Font metrics per glyph

See above- and below-anchor coordinates for A–Z, a–z, and IPA letters for Libertinus serif fonts (regular, italic, semibold, semibold italic) in four tables (Tables 1–4).

Table row headings

hex: 4-hex Unicode character codepoint.

reg, it, sb, si: regular, italic, semibold, semibold italic. In that row, the character is printed alone, with combining dot above, and with combining macron below, to show any anchor positioning. If no above- or below-anchor exists for that character, the character and combining mark are positioned by HarfBuzz fallback shaping.

ax, ay: X and Y coordinates of the above-anchor.

bx, by: X and Y coordinates of the below-anchor.

The next few rows use the bounding box of the glyph, because the midpoint of the width of the bounding box is a naive candidate for the ax and bx, and the bounding box coordinates are easy to determine.

bbm: midpoint of the bounding box X coordinate, $(X_{\text{min}} + X_{\text{max}})/2$, where the bounding box is characterized by $(X_{\text{min}}, Y_{\text{min}})$ and $(X_{\text{max}}, Y_{\text{max}})$.

$abb\delta$: X coordinate offset of above-anchor from midpoint of bounding box, $ax - bbm$.

$abb\%$: normalized offset, $abb\delta$ divided by bounding box width, $abb\delta/(X_{\text{max}} - X_{\text{min}})$.

$bbb\delta$: X coordinate offset of below-anchor from midpoint of bounding box, $bx - bbm$.

$bbb\%$: normalized offset, $bbb\delta$ divided by bounding box width, $bbb\delta/(X_{\text{max}} - X_{\text{min}})$.

There are better ways to define the geometric horizontal center of a glyph than the bounding-box midpoint, depending on the outline at the top or bottom of the glyph.

Vertical aspects, anchor Y coordinates, and clearance

A font has at least five vertical aspects, where glyph outlines live inside: descender height, baseline, x-height, ascender height, capital height. The values or ranges of these Y coordinates are in the first five rows of Table 5. These vertical aspects are the gray horizontal lines of Figure 1. These vertical aspects are displayed in a FontGorge glyph outline window as black horizontal guidelines with red shaded bands for ranges. I will label these as yra, yrc, yrx, $yr0$, yrd because these are Y values, and r indicates a reference value. a, c, x, 0, d indicate the five vertical aspects.

In the anchor model for Libertinus, the Y coordinate of an above-anchor or a below-anchor is at a common value depending on where the glyph outline lives relative to a vertical aspect. See the rows for ay and by in Tables 1–4 and find the common values according to the glyph's vertical aspects. There is no set name in font design for these above- or below-anchor Y coordinate reference values by ascender aspect, so I will call them $ayra$, $ayrc$, $ayrx$, $byr0$, $byrd$, These are the middle five rows of Table 5 For example, all glyphs that sit on the baseline have below-anchor Y coordinate $by = -110$, and thus $byr0 = -110$. All capitals A–Z have $ay = 850$, and thus $ayrc = 850$. Any any glyph that descends to the descender value or range $byrd$ should have $by = byrd$.

The clearance is defined as the difference between the anchor reference value and the vertical aspect Y value. I will label these clearances as $ay\delta a$, $ay\delta c$, $ay\delta x$, $by\delta 0$, $by\delta d$, For example, capital letters A–Z have a bounding box with max Y at 645, and all capital letters A–Z have an above anchor at $Y = 885$. So the above-anchor capital clearance is $ay\delta c = 885 - 645 = 205$. Because of the weird anchor model for Libertinus, this clearance value is not the exact vertical distance between a base and a mark, but this clearance value is still useful.

A glyph may not be a full ascender or a full descender, and its bounding box may not live closely within two vertical aspects. In such a case, the anchor Y coordinate is set by looking for the nearest, or most suitable, vertical aspect, and considering the clearance of that vertical aspect relative to the bounding box of the glyph.

For example, see the calculation in a following section of the below-anchor for Q, whose tail does not descend as deep as the full descender of J. The clearance $byrd$ can be used to calculate by of Q, after considering Q's bounding box min Y coordinate, as well as the shape of Q's tail.

This is the heuristic algorithm for calculating the Y coordinate of the above- or below anchor (ay or by) of a glyph that is not a full ascender or descender.

Table 1: Font metrics for A–Z

hex	0041	0042	0043	0044	0045	0046	0047	0048	0049	004A	004B	004C	004D	004E	004F	0050	0051	0052	0053	0054	0055	0056	0057	0058	0059	005A
reg	AȦẠ	BḂḄ	CĊC̣	DḊḌ	EĖẸ	FḞḞ̣	GĠG̣	HḢḤ	IİỊ	JJ̇J̣	KḲḴ̣	LĿḶ	MṀṂ	NṄṆ	OȮỌ	PṖṢ	QQ̇Q̣	RṘṚ	SṠṢ	TṪṬ	UU̇Ụ	VV̇Ṿ	WẆẈ	XẊẌ	YẎỴ	ZŻẒ
ax	354	267	340	306	281	284	365	361	150	182	336	151	431	346	335	277	348	269	232	307	346	336	450	332	295	320
ay	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
bx	308	268	363	309	287	209	367	362	147	109	342	262	393	348	350	240		258	234	300	328	318	479	307	295	287
by	-110	-110	-110	-110	-110	-110	-110	-110	-110	-319	-110	-110	-110	-110	-110	-110		-110	-110	-110	-110	-110	-110	-110	-110	-110
□xm	346	280	324	336	271	244	351	362	148	120	323	260	418	353	351	259	359	301	235	297	331	325	474	323	287	309
axδ	8	-13	16	-30	10	39	13	-1	2	62	13	-109	13	-7	-16	17	-11	-32	-3	10	15	10	-24	9	7	10
ax%	0.01	-0.03	0.03	-0.05	0.02	0.09	0.02	-0.00	0.01	0.16	0.02	-0.22	0.02	-0.01	-0.03	0.04	-0.02	-0.06	-0.01	0.02	0.02	0.02	-0.03	0.01	0.01	0.02
bxδ	-38	-12	39	-27	16	-35	15	0	-1	-11	19	1	-25	-5	-1	-19		-43	-1	3	-3	-7	5	-16	7	-22
bx%	-0.06	-0.02	0.07	-0.04	0.03	-0.08	0.02	0.00	-0.00	-0.03	0.03	0.00	-0.03	-0.01	-0.00	-0.04		-0.07	-0.00	0.01	-0.00	-0.01	0.01	-0.02	0.01	-0.04
it	*AȦẠ*	*BḂḄ*	*CĊC̣*	*DḊḌ*	*EĖẸ*	*FḞḞ̣*	*GĠG̣*	*HḢḤ*	*IİỊ*	*JJ̇J̣*	*KḲḴ̣*	*LĿḶ*	*MṀṂ*	*NṄṆ*	*OȮỌ*	*PṖṢ*	*QQ̇Q̣*	*RṘṚ*	*SṠṢ*	*TṪṬ*	*UU̇Ụ*	*VV̇Ṿ*	*WẆẈ*	*XẊẌ*	*YẎỴ*	*ZŻẒ*
ax	505	447	500	396	413	406	513	481	316	392	466	389	557	476	515	426	498	419	402	457	486	481	610	482	435	450
ay	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
bx	308	238	333	279	269	131	345	320	127	109	292	252	363	328	310	140		248	204	277	308	263	439	287	270	267
by	-108	-108	-110	-107	-107	-111	-110	-107	-109	-319	-110	-110	-110	-109	-113	-110		-111	-111	-109	-109	-110	-107	-107	-109	-107
□xm	334	289	379	355	282	282	395	398	203	263	370	268	445	400	396	295	396	295	272	383	441	418	587	367	393	357
axδ	170	158	120	41	131	123	118	83	113	128	95	120	112	76	118	131	101	124	130	73	45	63	23	114	42	93
ax%	0.26	0.29	0.21	0.06	0.24	0.23	0.19	0.11	0.31	0.28	0.14	0.24	0.13	0.10	0.19	0.23	0.16	0.22	0.29	0.13	0.07	0.10	0.03	0.16	0.08	0.15
bxδ	-26	-51	-46	-76	-13	-151	-50	-78	-76	-154	-78	-16	-82	-72	-86	-155		-47	-68	-106	-133	-155	-148	-80	-123	-90
bx%	-0.04	-0.09	-0.08	-0.11	-0.02	-0.28	-0.08	-0.10	-0.21	-0.33	-0.11	-0.03	-0.10	-0.09	-0.14	-0.28		-0.08	-0.15	-0.20	-0.21	-0.25	-0.16	-0.11	-0.23	-0.14
sb	**AȦẠ**	**BḂḄ**	**CĊC̣**	**DḊḌ**	**EĖẸ**	**FḞḞ̣**	**GĠG̣**	**HḢḤ**	**IİỊ**	**JJ̇J̣**	**KḲḴ̣**	**LĿḶ**	**MṀṂ**	**NṄṆ**	**OȮỌ**	**PṖṢ**	**QQ̇Q̣**	**RṘṚ**	**SṠṢ**	**TṪṬ**	**UU̇Ụ**	**VV̇Ṿ**	**WẆẈ**	**XẊẌ**	**YẎỴ**	**ZŻẒ**
ax										203					363		363	304	252	298		331	493	364	316	321
ay										806					802		802	808	802	802		805	805	804	805	803
bx								362		109								292	234	301		303	470	318	316	308
by								-110		-319								-111	-111	-107		-108	-107	-107	-109	-107
□xm	349	299	332	351	283	268	368	362	160	131	340	267	427	354	365	277	365	320	237	299	339	318	489	354	308	309
axδ										72					-2		-2	-16	14	-1		12	4	9	7	12
ax%										0.17					-0.00		-0.00	-0.03	0.04	-0.00		0.02	0.00	0.01	0.01	0.02
bxδ								0		-22								-28	-3	2		-15	-19	-36	7	-1
bx%								0.00		-0.05								-0.05	-0.01	0.00		-0.02	-0.02	-0.05	0.01	-0.00
si	***AȦẠ***	***BḂḄ***	***CĊC̣***	***DḊḌ***	***EĖẸ***	***FḞḞ̣***	***GĠG̣***	***HḢḤ***	***IİỊ***	***JJ̇J̣***	***KḲḴ̣***	***LĿḶ***	***MṀṂ***	***NṄṆ***	***OȮỌ***	***PṖṢ***	***QQ̇Q̣***	***RṘṚ***	***SṠṢ***	***TṪṬ***	***UU̇Ụ***	***VV̇Ṿ***	***WẆẈ***	***XẊẌ***	***YẎỴ***	***ZŻẒ***
ax	354	267	340	306	273	276	365	361	150	432	336	149	431	346	335	316	348	269	232	307	346	336	450	332	295	320
ay	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
bx	308	268	363	279	279	201	367	365	147	109	342	260	393	348	350	240		258	234	300	328	318	479	307	295	287
by	-110	-110	-110	-110	-110	-110	-110	-110	-110	-319	-110	-110	-110	-110	-110	-110		-110	-110	-110	-110	-110	-110	-110	-110	-110
□xm	364	298	378	367	305	304	405	445	244	288	421	312	449	452	435	307	435	309	263	381	470	417	581	387	407	350
axδ	-10	-31	-38	-61	-32	-28	-40	-84	-94	144	-85	-163	-18	-106	-100	8	-87	-40	-31	-74	-124	-81	-131	-55	-112	-30
ax%	-0.01	-0.05	-0.06	-0.09	-0.06	-0.05	-0.06	-0.11	-0.23	0.30	-0.12	-0.31	-0.02	-0.13	-0.15	0.01	-0.13	-0.07	-0.07	-0.13	-0.18	-0.13	-0.14	-0.07	-0.19	-0.05
bxδ	-56	-30	-15	-88	-26	-103	-38	-80	-97	-179	-79	-52	-56	-104	-85	-67		-51	-29	-81	-142	-99	-102	-80	-112	-63
bx%	-0.08	-0.05	-0.03	-0.13	-0.05	-0.18	-0.06	-0.10	-0.23	-0.37	-0.11	-0.10	-0.07	-0.13	-0.13	-0.12		-0.09	-0.07	-0.14	-0.21	-0.16	-0.11	-0.11	-0.19	-0.10

Table 2: Font metrics for a–z

hex	0061	0062	0063	0064	0065	0066	0067	0068	0069	006A	006B	006C	006D	006E	006F	0070	0071	0072	0073	0074	0075	0076	0077	0078	0079	007A
reg	aȧa̱	bḃḇ	cċc̱	dḋḍ	eėę	fḟf̱	gġg	hḣḥ	iii̱	jjj̱	kk̇ḵ	llḻ	mṁṃ	nṅṇ	oȯọ	pṗp	qq̇q	rṙṛ	sṡṣ	tṫṭ	uu̇ṳ	vv̇v̱	wẇw̱	xẋx̱	yẏy̱	zżẕ
ax	209		242	218	243		231	299	137		306	135	392	264	238	271	238	191	206		255	256	345	256	262	220
ay	645		645	645	645		645	645	795		784	885	645	646	645	646	645	646	645		645	646	646	646	645	646
bx	196	234	236	221	234	132	229	272	136	109	290	135	408	265	243	179	348	135	196	163	239	241	352	224	242	207
by	-110	-110	-110	-110	-110	-110	-319	-110	-110	-319	-214	-110	-110	-110	-110	-319	-319	-110	-110	-110	-110	-110	-110	-110	-319	-110
□xm	245	231	217	270	222	208	256	270	142	75	261	134	401	275	252	242	268	190	199	163	271	249	373	245	260	220
axδ	-36		24	-52	21		-25	28	-5		45	1	-9	-11	-14	28	-30	1	7		-16	7	-28	11	2	0
ax%	-0.09		0.07	-0.11	0.06		-0.06	0.06	-0.02		0.09	0.00	-0.01	-0.02	-0.03	0.06	-0.07	0.00	0.02		-0.03	0.01	-0.04	0.02	0.00	-0.00
bxδ	-49	2	18	-49	12	-76	-27	1	-6	33	29	1	6	-10	-9	-63	79	-55	-3	0	-32	-8	-21	-21	-18	-13
bx%	-0.12	0.01	0.05	-0.11	0.03	-0.21	-0.06	0.00	-0.03	0.14	0.06	0.00	0.01	-0.02	-0.02	-0.13	0.17	-0.16	-0.01	0.00	-0.06	-0.02	-0.03	-0.05	-0.04	-0.04
it	*aȧa̱*	*bḃḇ*	*cċc̱*	*dḋḍ*	*eėę*	*fḟf̱*	*gġg*	*hḣḥ*	*iii̱*	*jjj̱*	*kk̇ḵ*	*llḻ*	*mṁṃ*	*nṅṇ*	*oȯọ*	*pṗp*	*qq̇q*	*rṙṛ*	*sṡṣ*	*tṫṭ*	*uu̇ṳ*	*vv̇v̱*	*wẇw̱*	*xẋx̱*	*yẏy̱*	*zżẕ*
ax	341		343	218	332		360	409			405	292	522	394	359	411	369	312	327		386		348	366	367	360
ay	647		645	645	645		645	645			646	890	645	646	645	646	645	646	645		646		646	646	644	646
bx	174	214	163	221	158	222	197	229	136	0	223	115	382	235	186	248	169	113	169	133	226		355	224	368	187
by	-108	-108	-113	-107	-108	-111	-319	-111	-107	-319	-107	-107	-105	-111	-107	-110	-110	-110	-111	-107	-111		-110	-110	-107	-111
□xm	285	295	253	314	253	222	261	308	202	112	298	202	449	304	271	245	294	259	218	232	311	300	408	285	299	260
axδ	56		90	-96	78		98	100			107	90	73	89	87	166	75	53	109		75		-60	80	68	99
ax%	0.14		0.26	-0.21	0.23		0.19	0.23			0.24	0.41	0.11	0.21	0.23	0.31	0.18	0.16	0.34		0.17		-0.09	0.16	0.14	0.24
bxδ	-111	-81	-90	-93	-95	0	-64	-79	-66	-112	-75	-87	-67	-69	-85	3	-125	-146	-49	-99	-85		-53	-61	69	-73
bx%	-0.27	-0.21	-0.26	-0.20	-0.28	-0.00	-0.13	-0.18	-0.35	-0.29	-0.17	-0.40	-0.10	-0.16	-0.22	0.01	-0.29	-0.43	-0.15	-0.38	-0.19		-0.08	-0.12	0.14	-0.18
sb	**aȧa̱**	**bḃḇ**	**cċc̱**	**dḋḍ**	**eėę**	**fḟf̱**	**gġg**	**hḣḥ**	**iii̱**	**jjj̱**	**kk̇ḵ**	**llḻ**	**mṁṃ**	**nṅṇ**	**oȯọ**	**pṗp**	**qq̇q**	**rṙṛ**	**sṡṣ**	**tṫṭ**	**uu̇ṳ**	**vv̇v̱**	**wẇw̱**	**xẋx̱**	**yẏy̱**	**zżẕ**
ax							231		137			142	396	264							264					
ay							645		795			885	645	646							646					
bx							229	277	136	109		135	396	265						163	248					
by							-319	-110	-107	-319		-110	-105	-111						-107	-111					
□xm	255	235	224	273	232	225	263	275	155	78	261	148	408	289	259	254	273	212	201	168	275	262	384	260	264	221
axδ							-32		-18			-6	-12	-25							-11					
ax%							-0.07		-0.08			-0.03	-0.02	-0.05							-0.02					
bxδ							-34	2	-19	31		-13	-12	-24						-5	-27					
bx%							-0.08	0.00	-0.08	0.11		-0.05	-0.02	-0.05						-0.02	-0.05					
si	***aȧa̱***	***bḃḇ***	***cċc̱***	***dḋḍ***	***eėę***	***fḟf̱***	***gġg***	***hḣḥ***	***iii̱***	***jjj̱***	***kk̇ḵ***	***llḻ***	***mṁṃ***	***nṅṇ***	***oȯọ***	***pṗp***	***qq̇q***	***rṙṛ***	***sṡṣ***	***tṫṭ***	***uu̇ṳ***	***vv̇v̱***	***wẇw̱***	***xẋx̱***	***yẏy̱***	***zżẕ***
ax	379		242	218	243		291	469	137		306	174	392	264	238	271	238	191	326		255	256	345		262	220
ay	645		645	645	645		645	645	795		784	890	645	646	645	646	645	646	645		645	646	646		645	646
bx	196	234	236	221	234	132	199	272	136	67	290	135	400	265	243	179	348	135	196	133	239	241	352		242	207
by	-110	-110	-110	-110	-110	-110	-319	-110	-110	-319	-214	-110	-110	-110	-110	-319	-319	-110	-110	-107	-110	-110	-110		-319	-110
□xm	317	303	277	336	277	219	270	338	221	132	339	226	473	344	294	257	313	295	245	259	324	311	418	297	282	267
axδ	61		-35	-118	-34		21	130	-84		-33	-52	-81	-80	-56	13	-75	-104	80		-69	-55	-73		-20	-47
ax%	0.13		-0.09	-0.22	-0.09		0.04	0.26	-0.38		-0.07	-0.20	-0.11	-0.16	-0.13	0.02	-0.16	-0.25	0.22		-0.14	-0.13	-0.12		-0.04	-0.11
bxδ	-121	-69	-41	-115	-43	-87	-71	-66	-85	-65	-49	-91	-73	-79	-51	-78	35	-160	-49	-126	-85	-70	-66		-40	-60
bx%	-0.25	-0.14	-0.10	-0.22	-0.11	-0.14	-0.13	-0.13	-0.38	-0.15	-0.10	-0.35	-0.10	-0.16	-0.12	-0.13	0.07	-0.39	-0.14	-0.40	-0.17	-0.16	-0.11		-0.08	-0.14

Table 3: Font metrics for IPA, pt. 1/2

hex	00E6	014B	0153	0250	0251	0252	0253	0255	0256	0257	0258	0259	025B	025C	025E	0260	0261	0262	0263	0264	0265	0266	026B	026C	026D	026E	026F
reg	æǽǽ	ŋŋ́ŋ	œœ́œ	ɐɐ́ɐ	ɑɑ́ɑ	ɒɒ́ɒ	ɓɓ́ɓ	ɕɕ́ɕ	ɖɖ́ɖ	ɗɗ́ɗ	ɘɘ́ɘ	əə́ə	ɛɛ́ɛ	ɜɜ́ɜ	ɞɞ́ɞ	ɠɠ́ɠ	ɡɡ́ɡ	ɢɢ́ɢ	ɣɣ́ɣ	ɤɤ́ɤ	ɥɥ́ɥ	ɦɦ́ɦ	ɧɧ́ɧ	ɬɬ́ɬ	ɮɮ́ɮ	ɭɭ́ɭ	ɮɮ́ɮ	ɯɯ́ɯ
ax	350	264	405	228	228	264	480	209	218	218	204	195	209	194	239	209	239	265	259	248	268	299	209	209	209	209	407
ay	645	646	645	645	645	645	630	645	645	645	645	645	645	645	645	645	645	645	645	645	645	565	645	645	645	645	645
bx	327	265		229	212	259	212	212	212	212	212	186	212	184	256	236	226	248	251	216	219	240	146	218	196	196	410
by	-109	-111		-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-110	-320	-110	-277	-114	-129	-114	-114	-120	-110	-110	-114
□xm	343	240	387	216	272	247	265	222	314	322	220	232	201	195	241	320	228	259	258	221	271	262	144	182	182	250	390
axδ	6	24	17	11	-44	17	215	-13	-96	-104	-16	-37	7	-1	-2	-111	11	6	1	27	-3	36	65	27	27	-41	16
ax%	0.01	0.06	0.02	0.03	-0.09	0.04	0.56	-0.04	-0.17	-0.18	-0.04	-0.10	0.02	-0.00	-0.01	-0.20	0.03	0.01	0.00	0.06	-0.01	0.07	0.23	0.08	0.09	-0.09	0.02
bxδ	-16	25		12	-60	12	-53	-10	-102	-110	-8	-46	10	-11	14	-84	-2	-11	-7	-5	-52	-22	2	36	14	-54	19
bx%	-0.03	0.06		0.03	-0.13	0.03	-0.14	-0.03	-0.19	-0.19	-0.02	-0.12	0.03	-0.04	0.03	-0.15	-0.01	-0.02	-0.01	-0.01	-0.10	-0.05	0.01	0.11	0.04	-0.12	0.03
it	*æǽǽ*	*ŋŋ́ŋ*	*œœ́œ*	*ɐɐ́ɐ*	*ɑɑ́ɑ*	*ɒɒ́ɒ*	*ɓɓ́ɓ*	*ɕɕ́ɕ*	*ɖɖ́ɖ*	*ɗɗ́ɗ*	*ɘɘ́ɘ*	*əə́ə*	*ɛɛ́ɛ*	*ɜɜ́ɜ*	*ɞɞ́ɞ*	*ɠɠ́ɠ*	*ɡɡ́ɡ*	*ɢɢ́ɢ*	*ɣɣ́ɣ*	*ɤɤ́ɤ*	*ɥɥ́ɥ*	*ɦɦ́ɦ*	*ɧɧ́ɧ*	*ɬɬ́ɬ*	*ɮɮ́ɮ*	*ɭɭ́ɭ*	*ɮɮ́ɮ*	*ɯɯ́ɯ*
ax	480	264	480						218	218	204		322					265			258	299					407
ay	645	646	645						645	645	645		654					645			645	565					645
bx	327	265		229	212	243	212	212	212	212	212	212	212	184	256			248		177	219	240	146	218			410
by	-109	-111		-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114			-114		-114	-129	-114	-114	-120			-114
□xm	394	291	399	299	297	303	286	253	325	396	260	268	252	228	292	402	279	291	346	297	313	309	210	240	193	284	472
axδ	85	-27	81						-107	-178	-56		69					-26			-55	-10					-65
ax%	0.13	-0.07	0.13						-0.21	-0.28	-0.16		0.19					-0.06			-0.13	-0.02					-0.09
bxδ	-67	-26		-70	-85	-60	-74	-41	-113	-184	-48	-56	-40	-44	-36			-43		-120	-94	-69	-64	-22			-62
bx%	-0.10	-0.07		-0.17	-0.19	-0.14	-0.17	-0.11	-0.23	-0.29	-0.14	-0.16	-0.11	-0.12	-0.08			-0.10		-0.34	-0.22	-0.16	-0.22	-0.07			-0.09
sb	**æǽǽ**	**ŋŋ́ŋ**	**œœ́œ**	**ɐɐ́ɐ**	**ɑɑ́ɑ**	**ɒɒ́ɒ**	**ɓɓ́ɓ**	**ɕɕ́ɕ**	**ɖɖ́ɖ**	**ɗɗ́ɗ**	**ɘɘ́ɘ**	**əə́ə**	**ɛɛ́ɛ**	**ɜɜ́ɜ**	**ɞɞ́ɞ**	**ɠɠ́ɠ**	**ɡɡ́ɡ**	**ɢɢ́ɢ**	**ɣɣ́ɣ**	**ɤɤ́ɤ**	**ɥɥ́ɥ**	**ɦɦ́ɦ**	**ɧɧ́ɧ**	**ɬɬ́ɬ**	**ɮɮ́ɮ**	**ɭɭ́ɭ**	**ɮɮ́ɮ**	**ɯɯ́ɯ**
ax	355		430										229														377
ay	645		645										645														645
bx				228								212											146	218			380
by				-114								-114											-114	-120			-114
□xm	362	248	401	207	276	251	268	232	322	323	230	228	212	215	249	336	235	265	272	202	277	274	156	190	193	273	354
axδ	-7		28										16														23
ax%	-0.01		0.04										0.05														0.03
bxδ				21								-16											-10	27			26
bx%				0.05								-0.04											-0.04	0.08			0.03
si	***æǽǽ***	***ŋŋ́ŋ***	***œœ́œ***	***ɐɐ́ɐ***	***ɑɑ́ɑ***	***ɒɒ́ɒ***	***ɓɓ́ɓ***	***ɕɕ́ɕ***	***ɖɖ́ɖ***	***ɗɗ́ɗ***	***ɘɘ́ɘ***	***əə́ə***	***ɛɛ́ɛ***	***ɜɜ́ɜ***	***ɞɞ́ɞ***	***ɠɠ́ɠ***	***ɡɡ́ɡ***	***ɢɢ́ɢ***	***ɣɣ́ɣ***	***ɤɤ́ɤ***	***ɥɥ́ɥ***	***ɦɦ́ɦ***	***ɧɧ́ɧ***	***ɬɬ́ɬ***	***ɮɮ́ɮ***	***ɭɭ́ɭ***	***ɮɮ́ɮ***	***ɯɯ́ɯ***
ax	350	264	415		228	264			218	218	204		209	194				265			258	299					407
ay	645	646	645		645	645			645	645	645		645	645				645			645	565					645
bx	327	265		229	212	259	212	212	212	212	212		212	184	256			248		177	219	240	146	218			410
by	-109	-111		-114	-114	-114	-114	-114	-114	-114	-114		-114	-114	-114			-110		-114	-129	-114	-114	-120			-114
□xm	413	323	457	290	310	317	297	292	339	416	266	278	265	250	308	415	288	311	369	298	369	277	227	288	235	298	467
axδ	-63	-59	-42		-82	-53			-121	-198	-62		-56	-56				-46			-111	21					-60
ax%	-0.09	-0.13	-0.06		-0.17	-0.11			-0.22	-0.28	-0.14		-0.14	-0.14				-0.10			-0.21	0.04					-0.08
bxδ	-86	-58		-61	-98	-58	-85	-80	-127	-204	-54		-53	-66	-52			-63		-121	-150	-37	-81	-70			-57
bx%	-0.12	-0.12		-0.14	-0.20	-0.12	-0.18	-0.18	-0.24	-0.29	-0.12		-0.13	-0.17	-0.11			-0.13		-0.31	-0.29	-0.07	-0.23	-0.20			-0.08

Table 4: Font metrics for IPA, pt. 2/2

hex	0271	0272	0273	0274	0275	0276	0278	0279	027A	027B	027D	0280	0281	0282	0283	0288	028A	028B	028C	028E	028F	0290	0292	029B	029D	029F	03B2
reg	ɱɱ̀ɱ̀	ɲɲ̀ɲ̀	ɳɳ̀ɳ̀	ɴɴ̀ɴ̀	ɵɵ̀ɵ̀	ɶɶ̀ɶ̀	ɸɸ̀ɸ̀	ɾɾ̀ɾ̀	ɽɽ̀ɽ̀	ɻɻ̀ɻ̀	ʈʈ̀ʈ̀	ʀʀ̀ʀ̀	ʁʁ̀ʁ̀	ʂʂ̀ʂ̀	ʃʃ̀ʃ̀	ʧʧ̀ʧ̀	ʊʊ̀ʊ̀	ʊʊ̀ʊ̀	ʌʌ̀ʌ̀	ʎʎ̀ʎ̀	ʏʏ̀ʏ̀	ʐʐ̀ʐ̀	ʒʒ̀ʒ̀	ɡɡ̀ɡ̀	ɟɟ̀ɟ̀	ɭɭ̀ɭ̀	ββ̀β̀
ax	383	264	263	209	209	209	209	209	209		209	209	209	209	209	209	209	259	209	209	209	209	218	209	209	209	
ay	645	646	645	645	645	645	645	645	645		645	645	645	645	645	645	645	645	645	645	645	645	646	645	645	645	
bx	383	265	262	247	196	196	196	190	203	196	196	196	196	196	196	196	258	251	196	196	239	196	185	196	196	196	275
by	-105	-111	-112	-114	-110	-110	-110	-114	-114	-110	-110	-110	-110	-110	-110	-110	-114	-114	-110	-110	-107	-110	-319	-110	-110	-110	-114
□xm	365	234	295	278	252	340	285	177	179	225	186	238	238	199	140	157	253	235	244	248	236	262	216	294	126	207	261
axδ	17	29	-32	-69	-43	-131	-76	31	29		22	-29	-29	10	68	51	-44	24	-35	-39	-27	-53	1	-85	83	2	
ax%	0.03	0.05	-0.06	-0.14	-0.10	-0.21	-0.15	0.09	0.09		0.07	-0.07	-0.07	0.03	0.17	0.19	-0.09	0.06	-0.07	-0.08	-0.06	-0.12	0.00	-0.16	0.29	0.01	
bxδ	17	30	-33	-31	-56	-144	-89	12	23	-29	9	-42	-42	-3	55	38	4	16	-48	-52	3	-66	-31	-98	70	-11	13
bx%	0.03	0.06	-0.06	-0.06	-0.13	-0.23	-0.18	0.04	0.07	-0.07	0.03	-0.10	-0.10	-0.01	0.14	0.14	0.01	0.04	-0.10	-0.10	0.01	-0.15	-0.09	-0.19	0.24	-0.03	0.03
it	*ɱɱ̀ɱ̀*	*ɲɲ̀ɲ̀*	*ɳɳ̀ɳ̀*	*ɴɴ̀ɴ̀*	*ɵɵ̀ɵ̀*	*ɶɶ̀ɶ̀*	*ɸɸ̀ɸ̀*	*ɾɾ̀ɾ̀*	*ɽɽ̀ɽ̀*	*ɻɻ̀ɻ̀*	*ʈʈ̀ʈ̀*	*ʀʀ̀ʀ̀*	*ʁʁ̀ʁ̀*	*ʂʂ̀ʂ̀*	*ʃʃ̀ʃ̀*	*ʧʧ̀ʧ̀*	*ʊʊ̀ʊ̀*	*ʊʊ̀ʊ̀*	*ʌʌ̀ʌ̀*	*ʎʎ̀ʎ̀*	*ʏʏ̀ʏ̀*	*ʐʐ̀ʐ̀*	*ʒʒ̀ʒ̀*	*ɡɡ̀ɡ̀*	*ɟɟ̀ɟ̀*	*ɭɭ̀ɭ̀*	*ββ̀β̀*
ax	383	264	263															259					328				
ay	645	646	645															645					646				
bx	383	265	262	247				190	203								258	251			239		130				275
by	-105	-111	-112	-114				-114	-114								-114	-114			-107		-319				-114
□xm	388	210	295	328	289	376	342	98	268	233	265	245	260	217	197	224	312	323	248	305	332	251	236	361	145	222	261
axδ	-5	53	-32															-64					92				
ax%	-0.01	0.09	-0.08															-0.15					0.22				
bxδ	-5	54	-33	-81				92	-65								-54	-72			-93		-106				14
bx%	-0.01	0.09	-0.08	-0.14				0.27	-0.13								-0.11	-0.17			-0.21		-0.25				0.03
sb	**ɱɱ̀ɱ̀**	**ɲɲ̀ɲ̀**	**ɳɳ̀ɳ̀**	**ɴɴ̀ɴ̀**	**ɵɵ̀ɵ̀**	**ɶɶ̀ɶ̀**	**ɸɸ̀ɸ̀**	**ɾɾ̀ɾ̀**	**ɽɽ̀ɽ̀**	**ɻɻ̀ɻ̀**	**ʈʈ̀ʈ̀**	**ʀʀ̀ʀ̀**	**ʁʁ̀ʁ̀**	**ʂʂ̀ʂ̀**	**ʃʃ̀ʃ̀**	**ʧʧ̀ʧ̀**	**ʊʊ̀ʊ̀**	**ʊʊ̀ʊ̀**	**ʌʌ̀ʌ̀**	**ʎʎ̀ʎ̀**	**ʏʏ̀ʏ̀**	**ʐʐ̀ʐ̀**	**ʒʒ̀ʒ̀**	**ɡɡ̀ɡ̀**	**ɟɟ̀ɟ̀**	**ɭɭ̀ɭ̀**	**ββ̀β̀**
ax																	266						234				
ay																	645						645				
bx					229			190															193				
by					-114			-114															-319				
□xm	363	245	306	290	259	334	297	144	205	237	198	235	220	201	152	182	267	255	231	243	244	254	227	290	132	219	290
axδ																	-1						7				
ax%																	-0.00						0.02				
bxδ					-30			45															-34				
bx%					-0.07			0.13															-0.08				
si	***ɱɱ̀ɱ̀***	***ɲɲ̀ɲ̀***	***ɳɳ̀ɳ̀***	***ɴɴ̀ɴ̀***	***ɵɵ̀ɵ̀***	***ɶɶ̀ɶ̀***	***ɸɸ̀ɸ̀***	***ɾɾ̀ɾ̀***	***ɽɽ̀ɽ̀***	***ɻɻ̀ɻ̀***	***ʈʈ̀ʈ̀***	***ʀʀ̀ʀ̀***	***ʁʁ̀ʁ̀***	***ʂʂ̀ʂ̀***	***ʃʃ̀ʃ̀***	***ʧʧ̀ʧ̀***	***ʊʊ̀ʊ̀***	***ʊʊ̀ʊ̀***	***ʌʌ̀ʌ̀***	***ʎʎ̀ʎ̀***	***ʏʏ̀ʏ̀***	***ʐʐ̀ʐ̀***	***ʒʒ̀ʒ̀***	***ɡɡ̀ɡ̀***	***ɟɟ̀ɟ̀***	***ɭɭ̀ɭ̀***	***ββ̀β̀***
ax	383	264	263															259					218				
ay	645	646	645															645					646				
bx	383	265	262	247				190	203								258	251			239		141				275
by	-105	-111	-112	-114				-114	-114								-114	-114			-107		-319				-114
□xm	401	227	321	340	321	389	347	225	280	246	272	257	274	244	209	237	325	336	241	303	341	281	237	390	159	235	312
axδ	-18	37	-58															-77					-19				
ax%	-0.02	0.06	-0.09															-0.17					-0.04				
bxδ	-18	38	-59	-93				-35	-77								-67	-85			-102		-96				-37
bx%	-0.02	0.06	-0.10	-0.15				-0.08	-0.15								-0.12	-0.19			-0.22		-0.19				-0.07

If ay is not set for a glyph, find the most nearest, or most suitable, vertical aspect – ascender a, capital c, or x-height x. Then $ay = Y + ay\delta[acx] - \epsilon$, where Y = max Y of the bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

If by is not set for a glyph, find the most nearest, or most suitable, vertical aspect – baselines 0 or desender d. Then $by = Y + by\delta[0d] + \epsilon$, where Y = min Y of the bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

Table 5: Vertical aspects, and anchor Y coordinates and clearances of Libertinus

	v. aspect	Regular	Italic	Semibold	Semibold Italic
yra	ascender	688 [688,698]	688 [688,698]	690	698
yrc	capital	645 [645,658]	645 [645,658]	645	645
yrx	x-height	429 [429,432]	429 [429,432]	433	434 [433,447]
$yr0$	baseline	0 [-12,0]	0 [-12,0]	0	0
yrd	descender	-232 [-238,-227]	-233 [-238,-227]	-238	-232 [-238,-232]
$ayra$	ascender	885	885–890	885	885–890
$ayrc$	capital	850	850	805	850
$ayrx$	x-height	645	645	645	645
$byr0$	baseline	-110	-110	-110	-110
$byrd$	descender	-319	-319	-319	-319
$ay\delta a$	ascender	187	190	195	194
$ay\delta c$	capital	205	205	205	205
$ay\delta x$	x-height	216	216	212	198
$by\delta 0$	baseline	110	110	110	110
$by\delta d$	descender	87	87	81	100

Base coverage

Libertinus serif regular has good above- and below-anchors base coverage for A–Z, a–z, and IPA letters, in the sense that most anchors are defined. But I will show later that some anchors are not set to good values. The base coverage gets worse for italic and semiibold. Observe that many IPA letters reuse $ax = 209$ and $bx = 196$. These are good anchors for the letter a, but these were just copied to some IPA letters without regard to the letterform and are therefore bad anchors for those IPA letters.

Patch J and Q regular, below-anchor

J's below-anchor is too low. In Table 1, the Y coordinate of the below-anchor of J is the same as that of the lowercase letters with descenders. But the descender of J is not as low as that of j, p, q etc. Look at the J column in the grid labeled regular, Latin, below, which is later in this document. The below-marks are set with too much clearance below J.

Q has no below-anchor (Table 1). By harfbuzz fallback shaping (in the absence of anchors), a below-marks will be place below the baseline, without considering descenders. Look at the J column in the grid labeled regular, Latin, below, which is later in this document. All the below-marks crash into Q's descender.

Let's analyze the below-anchors in order to set the below-anchor for J and Q. The below clearance is clearly 110 units on the Y axis, as can be seen from the below-anchor Y coordinate of the letters that sit on the baseline. What should the below-anchor Y coordinate be for J and Q? For the Y coordinate, the lower bound of the J bounding box is at $Y = -172$. So, the below-anchor Y coordinate of J should arguably be $Y = -172 - 110 = -282$. But the designer set it at $Y = -319$, which is too low. For the Y coordinate, the lower bound of the Q bounding box is at $Y = -209$. So, the below-anchor Y coordinate of Q should arguably be $Y = -209 - 110 = -319$. But the J and Q have curves at the bottoms of their descenders, and

to account for overshoot, the anchor should be moved up by a few units. But what is the proper overshoot correction? The O has a baseline overshoot of 10 units. So, the below-anchor Y coordinate of J should really be $Y = -282 + 10 = -272$, and the below-anchor Y coordinate of Q should really be $Y = -319 + 10 = -309$.

Now, let's decide below-anchor X coordinate of Q. The Q and O are drawn with the same main oval outline coordinates. So, the below-anchor X coordinate of Q should the same as that for O, which is $X = 350$.

That is, the below-anchor of J should be $(109, -272)$, and the below-anchor of Q should be $(350, -309)$. Here is this positioning of J and Q in a patched version of the font (woth O and X for comparison only).

Original: X̯J̯Q̯O̯

Patched: X̯J̯Q̯O̯

But there is no sane reason to put a below-mark on J or Q, so this is not really a problem at all.

Patch k regular, below-anchor, Y coordinate

The below-anchor of k, regular, is at $Y = -214$. But it should be at $Y = -110$, like all other letters that sit on the baseline. That is, the below-anchor of k should be $(109, -110)$.

Here is this positioning of k in a patched version of the font (with o for comparison only).

Original: k̯ k̯o ok̯

Patched: k̯ k̯o ok̯

But there is no sane reason to put a below-mark on k, other than perhaps dot (which is a precomposed character), so this is not really a problem at all.

Above anchor, a–z regular

b, f, j, and t are missing the above-anchor. d and h have an above-anchor above the meanline but not above the ascender. l has an above-anchor above the ascender. k has an above-anchor, but it is just below the ascender, so any marks other than the do crash into the ascender. These are all legitimate design choices. There are no above-marks, other than those with precomposed characters, the would ever be combined with b, f, or t.

Patch j dot removal for three above-marks

The dot of i and j should be removed before adding any above-mark. But the dot is not removed for three above-marks: x, turned tilde, double macron. So the GSUB table should be patched to fir the dotless j substitution in this context.

To do.

Capital mark alternates

Libertinus has custom glyphs for some marks (acute, grave, cicrumflex, breve, a.o.) that are designed better for capital letters. These alternates are flatter. GSUB lookups 23 and 24 implement the substitution. This substitution is not fired on some capital consonants (T, V, W, X) because these marks are not expected on these bases.

Libertinus has custom glyphs for some marks that are designed better for superscripts. These alternates are smaller. Regrettably?, no GSUB table implements their substitution.

Regular, above-anchor, capitals, X coordinate

Figure 3 (left panel) shows pairs of bases aligned at their above-anchor. If a capital letter has a bad X coordinate for its above-anchor, then it will look always off, right or left, in its black column relative to the gray letters that it overlays. Note that the above-anchor X coordinate is not in the geometric (horizontal) center of the glyph, by design. The above-anchor X

coordinate is always off center by a few UPM units, sometimes left and sometimes right. But it very hard to discern a few UPM units. See the right panel of Figure 3 for an illustration of increasingly bad (spoofed) X coordinates of N's above-anchor.

Note that the above-anchor of L is above the major (vertical) stem of the L. For E, F, P and R, the above-anchor is optically centered above the glyph, not above the major stem.

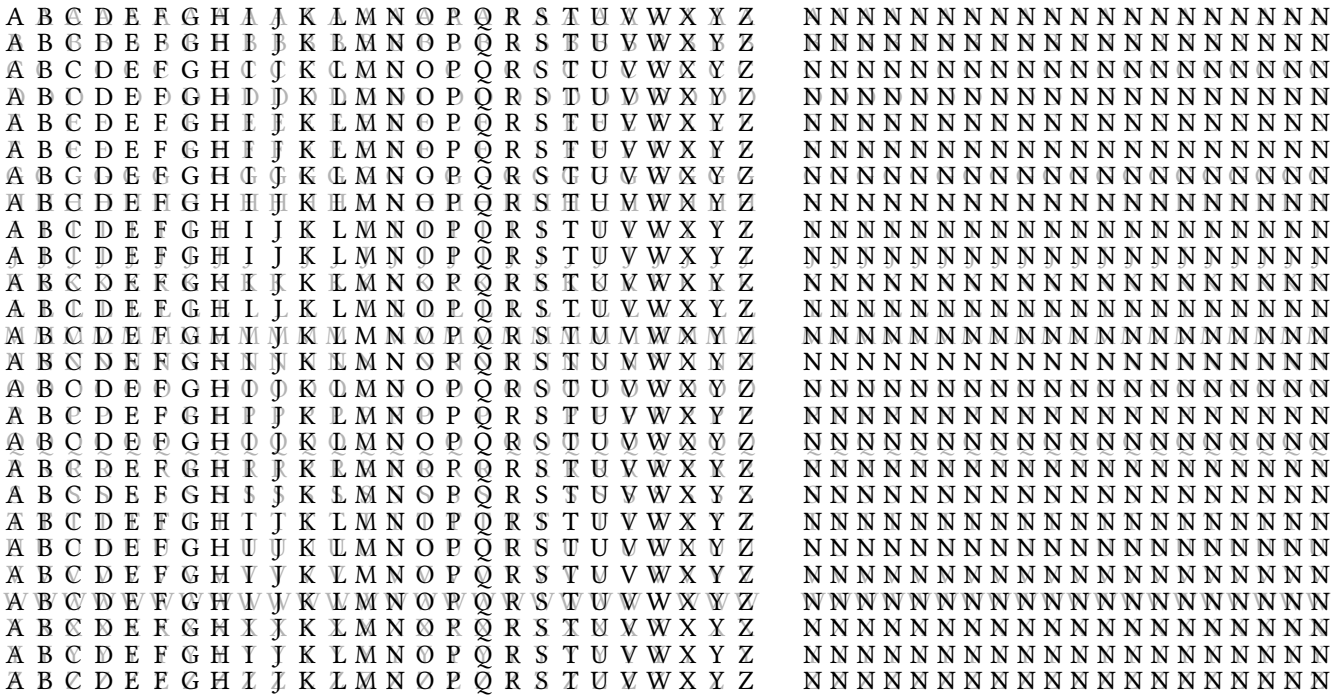

Figure 3: Above-anchor X coordinate alignment. In the left panel, the 26×26 grid shows letters A–Z aligned at the their above-anchor. The right panel shows how hard it is to discern bad X coordinates. The right panel shows N aligned with letters at their above-anchor, starting at the first column with N's above-anchor as currently set by the font, and then spoofing increasing bad values for the X coordinate of N's above-anchor, by 5 UPM per column. At the rightmost column, it becomes evident that the (spoofed) X coordinate of N's above-anchor is bad, because in that column the black Ns are too far right of the gray letters below them.

Comma-style marks

Libertinus has a some (precomposed) glyphs for Unicode characters (with code points) where the caron or cedilla has a comma-style shape, instead of a wedge or hook as might be naively expected. But all of these are appropriate for Czech/Slovak orthography (caron or *háček*) or Latvian orthography (cedilla, but really the comma-above mark or *komatiņš*, by a known mistake in Unicode).

Czech/Slovak *háček*

In Czech/Slovak orthography, the d, l, L, n and t may take a reduced caron that may look like a comma, not a wedge. In Libertinus, only the d, L, l, and t have the reduced comma-style caron in their precomposed form, and the n has wedge caron precomposed form.

`\char"010F` ď *ď* **ď** ***ď***

`\char"013D` Ľ *Ľ* **Ľ** ***Ľ***

`\char"013E` ľ *ľ* **ľ** ***ľ***

`\char"0148` ň *ň* **ň** ***ň***

`\char"0165` ť *ť* **ť** ***ť***

Libertinus even has a custom glyph for a small capital L with a caron, and it is comma-style.

`\textsc{l\char"030C}` ʟ̌

`\textsc{\char"013E}` ʟ̌

Cedilla and Latvian *komatiņš*

Here are all precomposed characters with a cedilla.

`\char"00C7` Ç *Ç* **Ç** ***Ç***	`\char"00E7` ç *ç* **ç** ***ç***
`\char"1E08` Ḉ *Ḉ* **Ḉ** ***Ḉ***	`\char"1E08` ḉ *ḉ* **ḉ** ***ḉ***
`\char"1E10` Ḑ *Ḑ* **Ḑ** ***Ḑ***	`\char"1E11` ḑ *ḑ* **ḑ** ***ḑ***
`\char"0228` Ȩ *Ȩ* **Ȩ** ***Ȩ***	`\char"0229` ȩ *ȩ* **ȩ** ***ȩ***
`\char"1E1C` Ḝ *Ḝ* **Ḝ** ***Ḝ***	`\char"1E1D` ḝ *ḝ* **ḝ** ***ḝ***
`\char"0122` Ģ *Ģ* **Ģ** ***Ģ***	`\char"0123` ġ *ġ* **ġ** ***ġ***
`\char"1E28` Ḩ *Ḩ* **Ḩ** ***Ḩ***	`\char"1E29` ḩ *ḩ* **ḩ** ***ḩ***
`\char"0136` Ķ *Ķ* **Ķ** ***Ķ***	`\char"0137` ķ *ķ* **ķ** ***ķ***
`\char"013B` Ļ *Ļ* **Ļ** ***Ļ***	`\char"013C` ļ *ļ* **ļ** ***ļ***
`\char"0145` Ņ *Ņ* **Ņ** ***Ņ***	`\char"0146` ņ *ņ* **ņ** ***ņ***
`\char"0156` Ŗ *Ŗ* **Ŗ** ***Ŗ***	`\char"0157` ŗ *ŗ* **ŗ** ***ŗ***
`\char"015E` Ş *Ş* **Ş** ***Ş***	`\char"015F` ş *ş* **ş** ***ş***
`\char"0162` Ţ *Ţ* **Ţ** ***Ţ***	`\char"0163` ţ *ţ* **ţ** ***ţ***

These characters – Ģ ġ Ķ ķ Ļ ļ Ņ ņ Ŗ ŗ – use the Latvian *komatiņš* which is proper for Latvian orthography, but the result of a historical mistake in Unicode naming.

The Unicode characters of D and d with cedilla exist for scholarly and historical reasons, not Latvian orthography. Fonts design these Unicode characters differently. Libertinus designs them inconsistently: comma-style in regular and italic, and hook-style in (semi)bold and (semi)bold italic. I think the cedilla on a D or d should be hook-like, but I do not really know.

The other characters – c, e, h, s, t – with a cedilla should definitely be hook-like, and they are so in Libertinus.

Somewhat mysteriously, if the LaTeX source tries to combine any of these letters – c, d, e, g, h, k, l, n, r, s, t – with the combining cedilla, U+0327, then the precomposed character is produced.

`c\char"0327` ç *ç* **ç** ***ç***
`d\char"0327` ḑ *ḑ* **ḑ** ***ḑ***
`e\char"0327` ȩ *ȩ* **ȩ** ***ȩ***
`g\char"0327` ġ *ġ* **ġ** ***ġ***
`h\char"0327` ḫ *ḫ* **ḫ** ***ḫ***
`k\char"0327` ķ *ķ* **ķ** ***ķ***
`l\char"0327` ļ *ļ* **ļ** ***ļ***
`n\char"0327` ṇ *ṇ* **ṇ** ***ṇ***
`r\char"0327` ŗ *ŗ* **ŗ** ***ŗ***
`s\char"0327` ş *ş* **ş** ***ş***
`t\char"0327` ţ *ţ* **ţ** ***ţ***

This behavior does not make sense for the Latvian *komatiņš* bases – g, k, l, n, r – because any person trying to produce Latvian g with *komatiņš* would not try `g\char"0327` or `\c{g}`. This behavior is tolerable for the other bases – c, d, e, h, s, t.

IJ

The ij and IJ ligatures are Unicode characters (with code points) and are supported by Libertinus.

IJ `\char"0132` U+0132 capital ligature IJ

ij `\char"0133` U+0133 small ligature ij

Libertinus has custom glyphs for these ligatures with an acute, and the small capital ligature with an acute. The GSUB table that activates these custom glyphs only works when the language is set to Dutch by a font loading command. For example,

```
\newfontfamily\SerifRegularDutch[Language = Dutch]{LibertinusSerif-Regular.otf}
```

ÍJ́ `{\SerifRegularDutch \char"0132\char"0301}` Dutch capital ligature IJ with acute

íj `{\SerifRegularDutch \char"0133\char"0301}` Dutch small ligature ij with acute

íj `{\SerifRegularDutch \textsc{\char"0133\char"0301}}` Dutch small capital ligature IJ with acute

In an English LaTeX document, `ij` and `IJ` are kerned by the GPOS table to produce ij and IJ, so they look close together, but they are not substituted by the single glyph for the ligature.

Tilde below, double acute above, double grave above, on vowels, in italic and semibold italic

Consider the tilde below. There is the combining tilde below, U+0330, encoded in LaTeX as the codepoint `\char"0330`. Remember to put the base before the combining mark in LaTeX, e.g. `a\char"0330` a̰. Three vowels (e, i, u) have precomposed characters in lowercase and capital, encoded in LaTeX as these codepoints:

`\char"1E1A \char"1E1B \char"1E2C \char"1E2D \char"1E74 \char"1E75` Ḛ ḛ Ḭ ḭ Ṵ ṵ *Ḛ ḛ Ḭ ḭ Ṵ ṵ* **Ḛ ḛ Ḭ ḭ Ṵ ṵ** ***Ḛ ḛ Ḭ ḭ Ṵ ṵ***

E, e, I, i, U, or u followed by `\char"0330` will produce the precomposed character glyph. But A, a, O, o, or any other vowel followed by `\char"0330` will be shaped by attaching the combining below tilde to that vowel at their below-anchor. Note the specific case of i. `i\char"0330 = \char"1E2D` ḭ = ḭ is a dotted i (a precomposed glyph).

Suppose you want a vowel with a tilde below, possibly also with a mark above, especially acute, grave, caron, double acute, or double grave. In the specific case of i, the above mark will crash into the dotted i unless you take care to encode the triplet starting with a dotless i, U+0131. For example, consider the combining acute U+0301.

`i\char"0330\char"0301 = \char"1E2D\char"0301` ḭ *ḭ* **ḭ** ***ḭ*** = ḭ *ḭ* **ḭ** ***ḭ***

`\char"0131\char"0330\char"0301` ı̰ *ı̰* **ı̰** ***ı̰***

But note that the position of the combining tilde below is bad in italic and semibold italic. This is because the below-anchor for the combining tilde below mark is bad. For tilde below, the precomposed character glyphs with i, e, and u are sometimes good and sometimes bad in italic and semibold italic.

`\char"00EC\char"0330` ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀***

`\char"0131\char"0300\char"0330` ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀***

`a\char"0330\char"030B` ấ̰ *ấ̰* **ấ̰** ***ấ̰***

`a\char"0330` ạ̰ *ạ̰* **ạ̰** ***ạ̰***

Patched

`\char"00EC\char"0330` ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀***

`\char"0131\char"0300\char"0330` ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀***

`a\char"0330\char"030B` ấ̰ *ấ̰* **ấ̰** ***ấ̰***

`a\char"0330` ạ̰ *ạ̰* **ạ̰** ***ạ̰***

`a\char"030B` ấ *ấ* **ấ** ***ấ***

f with dot below

tiɣə́ ṛf̣a

tiɣə́ ṛf̣a

tiɣə́ ṛf̣a

tiɣə́ ṛf̣a

ṛ = `\char"1E5B` is the precomposed character U+1E5B Latin small letter r with dot below.

f̣ = `f\char"0323` is f with U+0323 combining dot below.

Here are some letters with a dot below.

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

The left group of letters – a b d e h i k l m n o r s t u v w y z – have a precomposed character. For the right group of letters – c f g j p q x – the dot is placed according to the below-anchor (if the below-anchor is defined for the letter) or by harfbuzz fallback shaping (placing the dot below the baseline, regardless of the letter's bounding box).

In all styles – regular, italic, semibold, and semibold italic – the precomposed glyph of y with dot below places the dot beside the descender, even though the y has a below-anchor below the descender.

Do the letters c, f, g, j, p, q, and x have good below-anchors? In regular, the below-anchor of all these combining bases is below the glyph outline (good). In italic, the below-anchor of f, p, and q is beside the descender (bad), the below-anchor of g and j is below the descender (good), and the below-anchor of c and x is below the baseline (good). In semibold, the below-anchor of g and j is below the descender (good), but the other bases – c, f, p, q – have no below-anchor, and harfbuzz fallback positioning places the combining dot below the baseline, not below the bounding box; for c, f and x this is good, but for p and q this is bad, because of the descenders. In semibold italic, the below-anchor of c, g, j, p and q is below the glyph outline (good), the below-anchor of f is inside the descender (very bad), and x has no below-anchor but harfbuzz fallback positioning places the combining dot below the baseline (good).

Which dots are used in precomposed characters and how big are the dots? In all styles (except semibold italic), the precomposed characters reference the dot of `uni0307`, combining dot above, U+0307, and take care to translate this above-mark to a position below the letter outline, except for m, which references `dotbelowcomb`, combining dot below, U+0323. In semibold italic, `dotaccent` is used instead of `uni0307`. These two dots are four-point Bézier circles (in regular and semibold, but ovals in italic and semibold italic), but of different diameter. In

regular, `uni0307` has diameter 100, and `dotbelowcomb` has diameter 108. That is, the precomposed characters (except m) use a smaller dot, and the non-precomposed letters (and m) take a bigger dot. In semibold, the size difference is reversed: `uni0307` has larger diameter 124, and `dotbelowcomb` has small diameter 108. Actually, semibold `dotbelowcomb` is not a Bézier circle – the horizontal span is 109, and the vertical span is 108. In semibold italic, `dotaccent` is a skewed four-point Bézier oval – the horizontal and vertical spans are 120, offset by $(\pm 6, \pm 7.5)$ from the center — and `dotbelowcomb` is slightly larger, slightly flatter skewed six-point Bézier oval – horizontal span is 122 and the vertical span is 120. So, in semibold italic the two dots in precomposed and combining form are closest in size. Recall that f has a bad below-anchor, inside the descender, but g, j, p and q have a good below-anchor, below the glyph outline.

So, in order to solve the problem of the bad dot below f in italic, I need to decide the scope of the badness to correct. The simplest patch is move the below-anchor in italic and semibold italic below the descender. But this does not solve inconsistent below dot size across styles and letters. In italic and semibold italic, g and j have a (good) below-anchor at $Y = -319$, so that is also good Y coordinate for f's below-anchor.

I asked Copilot/GPT-5.1 to calculate a good X coordinate for f's below-anchor in italic and semibold italic, taking into account the U-pocket of f's descender and the below-anchor of `dotbelowcomb`. I decided that this U-pocket strategy was appropriate after testing other strategies that accounted for the slant of onf the major stroke. Here is a reasonbly faithful summary of the calculations and reasoning by Copilot/GPT-5.1.

> For italic, the descender's U-pocket is defined by the control points at $(14, -238)$ and $(33, -205)$, giving a U-center around $x \approx 24$. Using the (symmetric) italic `dotbelowcomb` anchor as the dot's optical center, the dot should sit under this U-pocket, not under the ball or the stem. A balanced, U-centered, slightly inner-biased choice is $X \approx 30$.
>
> For semibold italic, the U-pocket of the semibold italic descender is wider and more asymmetric, with a geometric U-center of $X = -33.5$. The semibold italic `dotbelowcomb` itself is also skewed, with its anchor already shifted relative to its mass. In practice, the visually correct position for the dot below f is under the open side of the U-pocket, clearly right of the ball and closer the right of the stem. The anchor candidate is $X \approx 62$.

Here are examples with the patched fonts, with f's below-anchor in italic at $(30, -319)$ and semibold italic at $(62, -319)$. This does not solve the problem of inconsistent dot size below letters, and I did not touch p, q, or y.

Original	Patched italics (f only)
tiɣə́ ṛfa	tiɣə́ ṛfa
tiɣə́ ṛfa	*tiɣə́ ṛfa*
tiɣə́ ṛfa	**tiɣə́ ṛfa**
tiɣə́ ṛfa	***tiɣə́ ṛfa***

Original (repeated from above)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç f ḡ j ṗ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç f ḡ j ṗ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç f ḡ j ṗ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç f ḡ j ṗ q ẋ

Patched italics (f only)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç f ḡ j ṗ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç f ḡ j ṗ q ẋ

Marks

Above marks – GPOS anchor 0

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

Below marks – GPOS anchor 2

x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

Left angle – GPOS anchor 3

x᜾ x᜾ x᜾ x᜾

Below-right marks – GPOS anchor 4

x̪ x̫ x̬

x̪ x̫ x̬

x̪ x̫ x̬

x̪ x̫ x̬

Cedilla – GPOS anchor 5

x̧ x̧ x̧ x̧

Overlay marks – GPOS anchor 6

o͓ o͔ o͕ o͖ o͗

o͓ o͔ o͕ o͖ o͗

o͓ o͔ o͕ o͖ o͗

o͓ o͔ o͕ o͖ o͗

Double marks

$\underbrace{xx}$ $\overbrace{xx}$ $\overline{xx}$ $\underline{xx}$ $\overset{\sim}{xx}$ $\overset{\frown}{xx}$ $\underset{\rightarrow}{xx}$

$\underbrace{xx}$ $\overbrace{xx}$ $\overline{xx}$ $\underline{xx}$ $\overset{\sim}{xx}$ $\overset{\frown}{xx}$ $\underset{\rightarrow}{xx}$

$\check{x}x$ $\acute{x}x$ $\grave{x}x$ $\hat{x}x$ $\breve{x}x$ $\check{x}x$ $\ddot{x}x$

$x\boxtimes x$ $x\boxtimes x$ $x\boxtimes x$ $x\boxtimes x$ $x\boxtimes x$ $x\boxtimes x$ $x\boxtimes x$

Overline and low line

O+0305 combining overline and U+0332 combining low line are applied by GSUB lookups that substitute a custom-width line mark according to width classes. GSUB lookup 51 effectively classifies the glyphs by width, and GSUB lookups 52–82 provide the appropriate custom-width mark for the base width class. Semibold and semibold italic do not have any combining low lines. Custom-width line marks do not have anchors. They are positioned by harfbuzz fallback shaping. The overlines are above the ascender line, so they are above all bases. The low lines are below the baseline, but above the descender line, and so they cut though any descenders. If a new glyph is added, and you want to support custom-width line marks for it, then remember to classify the new glyph in GSUB lookup 51.

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Above-right marks – GPOS anchor 1

Libertinus has three combining marks with GPOS anchor 1 – U+031B horn, U+0315 comma above right, and U+0358 dot above right.

A few bases in Libertinus have this anchor (anchor 1), but none that possibly need it.

Horn

The Vietnamese alphabet has two horn vowels – Ơơ Ưư– and these vowels may take tonal marks. All these combinations are Unicode characters (with code points).

Ớ ơ Ư ư Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

Ớ ơ Ư ư Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

Ớ ơ Ư ư Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

Ớ ơ Ư ư Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

There is a Unicode character for the combining horn mark: ̛ `\char"031B` U+031B, but this is not used in practice. The Vietnamese vowels are usually entered in a LaTeX document as Unicode characters, also with `\usepackage[vietnamese]{babel}` for hyphenation and spacing.

Comma above right

There is a Unicode character for the combining comma above right mark: ̕ `\char"0315` U+0315.

U+0315 (comma above right) is used in rare cases when another comma-like mark does not fit.

In IPA, the glottalization diacritic mark uses U+0313 (comma above) as a combining mark. But when a glottalization mark is needed for a superscript consonant (1) or superscript consonant cluster (2), then the combining mark U+0315 (comma above right) is used instead, because its positioning is hopefully better than that of U+0313.

(1) *k̓wkt* [k̓^{wkx}t] ‘to scrape’ (Nuxalk)

(2) *p’ak* [p’^{kx}ak] ‘he/she cut it’ (Upper Chehalis)

In (1), the superscript consonant cluster is ^{wkx} and it is glottalized. In (2), the superscript consonant ^{x} is glottalized.

Because U+0315 is being used in superscript position on a smaller sized glyph, U+0315 should probably be smaller than U+0313, but it is not so in Libertinus.

Dot above right

There is a Unicode character for the combining dot above right mark: ̘ `\char"0358` U+0358 dot above right

U+0358 should be a small, above-right dot, used to mark palatalization or similar secondary articulations on superscript consonants, created for the same typographic reason U+0315 exists – the standard palatalization diacritic (U+0307, dot above) is too big and too high.

In Libertinus, U+0358 is a 104-diameter Bezier circle, and U+0307 is a smaller Bezier oval, inverting the desired size difference.

There are only a few IPA superscript letters that may serve as the possible host of U+0315 or U+0358 on superscript consonants or on superscript consonant clusters. In Libertinus, regrettably none of these characters has the anchor (anchor 1), so the mark (U+0315 and U+0358) is positioned by harfbuzz fallback shaping, which seems to place the mark at the right boundary of the glyph at a constant vertical aspect (effectively at the superscript x-height).

These IPA marks U+0315 and U+0358 should only be used in regular (not italic and not semibold). Regrettably, in Libertinus serif semibold, U+0358 looks like a grave, not a dot, and there is no U+0358 glyph in Libertinus serif semibold italic.

There are three comma/apostrophe glyphs that could possibly mark glottalization: U+0313 combining comma above, U+0315 combining comma above right, and U+02BC modifier letter apostrophe. In Libertinus, U+0315 has the worst appearance, even though U+0315 is intended for superscript letters. In Gentium Plus (SIL Global), U+0315 is tightly above-right the superscript letter, as expected.

Libertinus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [k̓ʷᵏˣ̌t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣ̒t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [k̓ʷᵏˣʼt]

Gentium Plus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [k̓ʷᵏˣ̌t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣ̒t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [k̓ʷᵏˣʼt]

I patched Libertinus to add anchors to those IPA superscript letters to support U+0315.

Patched Libertinus, selected IPA superscript letters with U+0315 and U+0358

ʰ ʱ ʲ ʳ ʴ ʵ ʶ ʷ ʸ ʹ ˢ ˣ ˺ ᵇ ᵏ ᵐ ᵖ ᵗ ᵚ ᵛ ᵝ ᵝ ᵝ ᵝ

ʰ ʱ ʲ ʳ ʴ ʵ ʶ ʷ ʸ ʹ ˢ ˣ ˺ ᵇ ᵏ ᵐ ᵖ ᵗ ᵚ ᵛ ᵝ ᵝ ᵝ ᵝ

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣ̒t]

Gentium Plus, selected IPA superscript letters with U+0315 and U+0358

ʰ ʱ ʲ ʳ ʴ ʵ ʶ ʷ ʸ ˠ ˡ ˢ ˣ ˺ ᵇ ᵏ ᵐ ᵖ ᵗ ᵚ ᵛ ᵝ ᵝ ᵝ ᵝ

ʰ ʱ ʲ ʳ ʴ ʵ ʶ ʷ ʸ ˠ ˡ ˢ ˣ ˺ ᵇ ᵏ ᵐ ᵖ ᵗ ᵚ ᵛ ᵝ ᵝ ᵝ ᵝ

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣʼt]

Here are the IPA superscript letters considered and their new anchor (anchor 1).

U+02B0: (347, 648), U+02B1: (339, 648), U+02B2: (193, 648), U+02B3: (284, 648), U+02B4: (274, 648), U+02B5: (319, 648), U+02B6: (308, 648), U+02B7: (497, 648), U+02B8: (368, 648), U+02E0: (341, 648), U+02E1: (193, 648), U+02E2: (270, 648), U+02E3: (344, 648), U+02E4: (282, 648), U+1D47: (319, 648), U+1D4F: (353, 648), U+1D50: (533, 648), U+1D56: (334, 648), U+1D57: (241, 648), U+1D5A: (525, 648), U+1D5B: (354, 648), U+1D5C: (508, 648), U+1D5D: (335, 648), U+1D5E: (372, 648), U+1D5F: (353, 648)

I did not change the anchor-1 of U+315 and U+0358. U+0358 has its anchor at the xMin and yMin of its bounding box. But U+0315 already has some clearance built in to its anchor: its anchor is 62 units below its yMin, but 25 units right of it xMin. So, I decided to provide x-clearance of 12 units and y-clearance of 80 units from the bases. All these IPA superscript letters already have a superscript meanline at $Y = 630$. So their anchor will be at $Y = 630 - 62 + 80 = 648$. The X coordinate depends on their bounding box and its xMax. To provide the desired clearance of 12 units, the anchor is at $X = \text{xMax} + 26 + 12$. These anchor coordinate values were optimized for U+0315 but they work fine for U+0358.

Bases

Latin

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſÐđĦħıȷŁłƀƁ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſÐđĦħıȷŁłƀƁ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſÐđĦħıȷŁłƀƁ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſÐđĦħıȷŁłƀƁ

IPA

ɒɔɕɖɗɑɤɜɘɛɜɞɟʃɡɠɣɤɥɦɧɨɪɭɬɮɭɫɱɲɳɴɵœɶɸɹɻʀʁʂʃʈʧʊʋʌɯʏɤʐʑʒʔʕʃʗʘʙɕʜʝʞʟꞵ
ɒɔɕɖɗɑɤɜɘɛɜɞɟʃɡɠɣɤɥɦɧɨɪɭɬɮɭɫɱɲɳɴɵœɶɸɹɻʀʁʂʃʈʧʊʋʌɯʏɤʐʑʒʔʕʃʗʘʙɕʜʝʞʟ□
ɒɔɕɖɗɑɤɜɘɛɜɞɟʃɡɠɣɤɥɦɧɨɪɭɬɮɭɫɱɲɳɴɵœɶɸɹɻʀʁʂʃʈʧʊʋʌɯʏɤʐʑʒʔʕʃʗʘʙɕʜʝʞʟ□
ɒɔɕɖɗɑɤɜɘɛɜɞɟʃɡɠɣɤɥɦɧɨɪɭɬɮɭɫɱɲɳɴɵœɶɸɹɻʀʁʂʃʈʧʊʋʌɯʏɤʐʑʒʔʕʃʗʘʙɕʜʝʞʟ□

Superscript consonants

ʰʱʲʳʴʵʶʷʸˡˢˣˤˢˤˢˤˢ

Precomposed capital vowels

ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẦẤẪẨẰẮẴẲẠẬẶẢẨẲẸỆỂỄỀẾỂỂỄỆỊỈỈĨỊỈĨỈỌỘỖỔỒỐỖỔỒỐỘỎỔỔỖỔỜỚỠỞỢỎỜỚỠỞỢỦỤỤŨ
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Precomposed small vowels

àáâãäåāăąǎǟǡǻȁȃầấẫẩằắẵẳạậặảẩẳẹệểễềếểểễệịỉỉĩịỉĩỉọộỗổồốỗổồốộỏổổỗổờớỡởợỏờớỡởợủụụũūŭůűũůűũůùùûũūŭůűũůűũůùùûũūŭůűũůűũůùùû

No marks: rare, digraph etc.

Ƹ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
Ƹ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
Ƹ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
Ƹ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj

Small capitals

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ

Precomposed characters of anchor-relevant bases and marks

à á â ã ā å ą ặ ằ ắ ẳ ấ ầ ẫ è é ê ẽ ē ę ệ ế ề ễ ì í î ĩ ī ị ò ó ô õ ō ọ ộ ố ồ ỗ ù ú û ũ ū ụ ỳ ý ỷ ỹ ȳ ỵ ḍ ḷ ṇ ṛ ṣ ṭ ẓ
à á â ã ā å ą ặ ằ ắ ẳ ấ ầ ẫ è é ê ẽ ē ę ệ ế ề ễ ì í î ĩ ī ị ò ó ô õ ō ọ ộ ố ồ ỗ ù ú û ũ ū ụ ỳ ý ỷ ỹ ȳ ỵ ḍ ḷ ṇ ṛ ṣ ṭ ẓ
à á â ã ā å ą ặ ằ ắ ẳ ấ ầ ẫ è é ê ẽ ē ę ệ ế ề ễ ì í î ĩ ī ị ò ó ô õ ō ọ ộ ố ồ ỗ ù ú û ũ ū ụ ỳ ý ỷ ỹ ȳ ỵ ḍ ḷ ṇ ṛ ṣ ṭ ẓ
à á â ã ā å ą ặ ằ ắ ẳ ấ ầ ẫ è é ê ẽ ē ę ệ ế ề ễ ì í î ĩ ī ị ò ó ô õ ō ọ ộ ố ồ ỗ ù ú û ũ ū ụ ỳ ý ỷ ỹ ȳ ỵ ḍ ḷ ṇ ṛ ṣ ṭ ẓ

Base and mark shaping

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic. Note that LSP uses the Libertinus fonts named “Semibold“, e.g. `LibertinusSerif-Semibold.otf`, for all boldface style.

A black color means GPOS anchor positioning.

A blue color means a precomposed glyph.

A red color means harfbuzz fallback shaping (no anchors, no precomposed glyph, no GSUB substitution).

A green color means that a GSUB substitution (to dotless i or j) was expected but did not occur.

A faded color – gray, light blue, pink, or light green – means that the base and mark combination is not reasonably expected in notation in linguistics.

A gray color can also mean that mark glyph is missing in the font. In this case, just the gray base is printed. This on happens in Libertinus Serif Semibold and Semibold Italic for three below marks: o̽ (U+0333 double macron below), o̪ (U+033A inverted bridge below), and o̺ (U+033B square below). Otherwise, all considered base and mark combinations can be drawn in XeLaTeX documents in Libertinus without tofu for missing glyphs.

There are separate grids for common marks (for IPA etc.) and rare marks (for UPA etc.). Each grid considers all possible bases for marks in that grid, and so the grid is actually a complete matrix or grid or table, but with faded colors indicating combinations that can be ignored because they are not expected in notation in linguistics.

Regular, Bases requiring anchors for common marks, Common combining marks

à ƀ ƈ ɖ è f ɡ ì j ƙ l̀ m̀ ǹ ò ƥ q̀ ɾ̀ s̀ t̀ ù ʋ̀ ẁ x̀ ỳ ƶ æ̀ œ̀ ŋ̀ ɛ̀ ɑ̀ ɔ̀ ɓ̀ ɗ̀ ɗ̀ ə̀ ə̀ ɛ̀ ɜ̀ ɛ̀ ɜ̀ ɡ̀ ɡ̀ ɡ̀ ɣ̀ ɣ̀ ɥ̀ ɦ̀ ɨ̀ ɨ̀ ɭ̀ ɮ̀ ʉ̀ ɯ̀ ɲ̀ ɳ̀ ɴ̀ ɵ̀ œ̀ ɪ̀ ɪ̀ ɻ̀ ʀ̀ ʁ̀ ʂ̀ ʃ̀ ʊ̀ ʌ̀ ʎ̀ ʏ̀ ʐ̀ ʑ̀ ʒ̀ ʝ̀ ɻ̀

á ƀ ć đ é f ɡ́ í j ƙ ĺ ḿ ń ó ƥ q́ ŕ ś ƭ ú ʋ́ ẃ x́ ý ź ǽ œ́ ŋ́ ɛ́ ɑ́ ɔ́ ɓ́ ɗ́ ɗ́ ə́ ə́ ɛ́ ɜ́ ɛ́ ɜ́ ɡ́ ɡ́ ɡ́ ɣ́ ɣ́ ɥ́ ɦ́ ɨ́ ɨ́ ɭ́ ɮ́ ʉ́ ɯ́ ɲ́ ɳ́ ɴ́ ɵ́ œ́ ɪ́ ɪ́ ɻ́ ʀ́ ʁ́ ʂ́ ʃ́ ʊ́ ʌ́ ʎ́ ʏ́ ʐ́ ʑ́ ʒ́ ʝ́ ɻ́

â ƀ ĉ đ ê f ɡ̂ î ĵ ƙ l̂ m̂ n̂ ô ƥ q̂ r̂ ŝ ƭ û ʋ̂ ŵ x̂ ŷ ẑ æ̂ œ̂ ŋ̂ ɛ̂ ɑ̂ ɔ̂ ɓ̂ ɗ̂ ɗ̂ ə̂ ə̂ ɛ̂ ɜ̂ ɛ̂ ɜ̂ ɡ̂ ɡ̂ ɡ̂ ɣ̂ ɣ̂ ɥ̂ ɦ̂ ɨ̂ ɨ̂ ɭ̂ ɮ̂ ʉ̂ ɯ̂ ɲ̂ ɳ̂ ɴ̂ ɵ̂ œ̂ ɪ̂ ɪ̂ ɻ̂ ʀ̂ ʁ̂ ʂ̂ ʃ̂ ʊ̂ ʌ̂ ʎ̂ ʏ̂ ʐ̂ ʑ̂ ʒ̂ ʝ̂ ɻ̂

ã ƀ c̃ đ ẽ f ɡ̃ ĩ j ƙ l̃ m̃ ñ õ ƥ q̃ r̃ s̃ ƭ ũ ʋ̃ w̃ x̃ ỹ z̃ æ̃ œ̃ ŋ̃ ɛ̃ ɑ̃ ɔ̃ ɓ̃ ɗ̃ ɗ̃ ə̃ ə̃ ɛ̃ ɜ̃ ɛ̃ ɜ̃ ɡ̃ ɡ̃ ɡ̃ ɣ̃ ɣ̃ ɥ̃ ɦ̃ ɨ̃ ɨ̃ ɭ̃ ɮ̃ ʉ̃ ɯ̃ ɲ̃ ɳ̃ ɴ̃ ɵ̃ œ̃ ɪ̃ ɪ̃ ɻ̃ ʀ̃ ʁ̃ ʂ̃ ʃ̃ ʊ̃ ʌ̃ ʎ̃ ʏ̃ ʐ̃ ʑ̃ ʒ̃ ʝ̃ ɻ̃

ā ƀ c̄ đ ē f ɡ̄ ī j ƙ l̄ m̄ n̄ ō ƥ q̄ r̄ s̄ ƭ ū ʋ̄ w̄ x̄ ȳ z̄ ǣ œ̄ ŋ̄ ɛ̄ ɑ̄ ɔ̄ ɓ̄ ɗ̄ ɗ̄ ə̄ ə̄ ɛ̄ ɜ̄ ɛ̄ ɜ̄ ɡ̄ ɡ̄ ɡ̄ ɣ̄ ɣ̄ ɥ̄ ɦ̄ ɨ̄ ɨ̄ ɭ̄ ɮ̄ ʉ̄ ɯ̄ ɲ̄ ɳ̄ ɴ̄ ɵ̄ œ̄ ɪ̄ ɪ̄ ɻ̄ ʀ̄ ʁ̄ ʂ̄ ʃ̄ ʊ̄ ʌ̄ ʎ̄ ʏ̄ ʐ̄ ʑ̄ ʒ̄ ʝ̄ ɻ̄

å ƀ c̊ đ ɛ̊ f ɡ̊ ɨ̊ j ƙ l̊ m̊ n̊ o̊ ƥ q̊ r̊ s̊ ƭ ů ʋ̊ ẘ x̊ ẙ z̊ æ̊ œ̊ ŋ̊ ɛ̊ ɑ̊ ɔ̊ ɓ̊ ɗ̊ ɗ̊ ə̊ ə̊ ɛ̊ ɜ̊ ɛ̊ ɜ̊ ɡ̊ ɡ̊ ɡ̊ ɣ̊ ɣ̊ ɥ̊ ɦ̊ ɨ̊ ɨ̊ ɭ̊ ɮ̊ ʉ̊ ɯ̊ ɲ̊ ɳ̊ ɴ̊ ɵ̊ œ̊ ɪ̊ ɪ̊ ɻ̊ ʀ̊ ʁ̊ ʂ̊ ʃ̊ ʊ̊ ʌ̊ ʎ̊ ʏ̊ ʐ̊ ʑ̊ ʒ̊ ʝ̊ ɻ̊

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

Regular patched, Bases requiring anchors for common marks, Common combining marks

à ƀ ƈ đ è f ɡ ì ɟ k̀ l̀ m̀ ǹ ò p̀ q̀ ɾ̀ s̀ t̀ ù v̀ ẁ x̀ ỳ ƶ æ̀ œ̀ ŋ̀ ɛ̀ ɑ̀ ɔ̀ ɓ̀ ɖ̀ ƒ̀ ə̀ ə̀ ɛ̀ ɜ̀ ɛ̀ ɟ̀ ɡ̀ ɡ̀ ɡ̀ ɣ̀ ɣ̀ ɥ̀ ɦ̀ ɨ̀ ɨ̀ ɭ̀ ɰ̀ ɯ̀ ɱ̀ ɲ̀ ɳ̀ ɴ̀ ɵ̀ œ̀ ɪ̀ ʏ̀ ɾ̀ ɹ̀ ʂ̀ ʃ̀ ʊ̀ ʌ̀ ʎ̀ ʏ̀ ʐ̀ ʑ̀ ʒ̀ ʝ̀ ɮ̀

á ƀ ć đ é f ɡ í ɟ ḱ ĺ ḿ ń ó ṕ q́ ɾ́ ś ť ú v́ ẃ x́ ý ź ǽ œ́ ŋ́ ɛ́ ɑ́ ɔ́ ɓ́ ɖ́ ƒ́ ə́ ə́ ɛ́ ɜ́ ɛ́ ɟ́ ɡ́ ɡ́ ɡ́ ɣ́ ɣ́ ɥ́ ɦ́ ɨ́ ɨ́ ɭ́ ɰ́ ɯ́ ɱ́ ɲ́ ɳ́ ɴ́ ɵ́ œ́ ɪ́ ʏ́ ɾ́ ɹ́ ʂ́ ʃ́ ʊ́ ʌ́ ʎ́ ʏ́ ʐ́ ʑ́ ʒ́ ʝ́ ɮ́

â ƀ ĉ đ ê f ɡ î ɟ k̂ l̂ m̂ n̂ ô p̂ q̂ ɾ̂ ŝ ť û v̂ ŵ x̂ ŷ ẑ æ̂ œ̂ ŋ̂ ɛ̂ ɑ̂ ɔ̂ ɓ̂ ɖ̂ ƒ̂ ə̂ ə̂ ɛ̂ ɜ̂ ɛ̂ ɟ̂ ɡ̂ ɡ̂ ɡ̂ ɣ̂ ɣ̂ ɥ̂ ɦ̂ ɨ̂ ɨ̂ ɭ̂ ɰ̂ ɯ̂ ɱ̂ ɲ̂ ɳ̂ ɴ̂ ɵ̂ œ̂ ɪ̂ ʏ̂ ɾ̂ ɹ̂ ʂ̂ ʃ̂ ʊ̂ ʌ̂ ʎ̂ ʏ̂ ʐ̂ ʑ̂ ʒ̂ ʝ̂ ɮ̂

ã ƀ ĉ đ ẽ f ɡ ĩ ɟ k̃ l̃ m̃ ñ õ p̃ q̃ ɾ̃ s̃ t̃ ũ ṽ w̃ x̃ ỹ z̃ æ̃ œ̃ ŋ̃ ɛ̃ ɑ̃ ɔ̃ ɓ̃ ɖ̃ ƒ̃ ə̃ ə̃ ɛ̃ ɜ̃ ɛ̃ ɟ̃ ɡ̃ ɡ̃ ɡ̃ ɣ̃ ɣ̃ ɥ̃ ɦ̃ ɨ̃ ɨ̃ ɭ̃ ɰ̃ ɯ̃ ɱ̃ ɲ̃ ɳ̃ ɴ̃ ɵ̃ œ̃ ɪ̃ ʏ̃ ɾ̃ ɹ̃ ʂ̃ ʃ̃ ʊ̃ ʌ̃ ʎ̃ ʏ̃ ʐ̃ ʑ̃ ʒ̃ ʝ̃ ɮ̃

ā ƀ ē đ ē f ɡ ī ɟ k̄ l̄ m̄ n̄ ō p̄ q̄ ɾ̄ s̄ t̄ ū v̄ w̄ x̄ ȳ z̄ ǣ œ̄ ŋ̄ ɛ̄ ɑ̄ ɔ̄ ɓ̄ ɖ̄ ƒ̄ ə̄ ə̄ ɛ̄ ɜ̄ ɛ̄ ɟ̄ ɡ̄ ɡ̄ ɡ̄ ɣ̄ ɣ̄ ɥ̄ ɦ̄ ɨ̄ ɨ̄ ɭ̄ ɰ̄ ɯ̄ ɱ̄ ɲ̄ ɳ̄ ɴ̄ ɵ̄ œ̄ ɪ̄ ʏ̄ ɾ̄ ɹ̄ ʂ̄ ʃ̄ ʊ̄ ʌ̄ ʎ̄ ʏ̄ ʐ̄ ʑ̄ ʒ̄ ʝ̄ ɮ̄

å ƀ ƈ đ ɛ̊ f ɡ ɨ̊ ɟ̊ k̊ l̊ m̊ n̊ ȯ p̊ q̊ ɾ̊ s̊ t̊ ů v̊ ẘ x̊ ẙ ƶ æ̊ œ̊ ŋ̊ ɛ̊ ɑ̊ ɔ̊ ɓ̊ ɖ̊ ƒ̊ ə̊ ə̊ ɛ̊ ɜ̊ ɛ̊ ɟ̊ ɡ̊ ɡ̊ ɡ̊ ɣ̊ ɣ̊ ɥ̊ ɦ̊ ɨ̊ ɨ̊ ɭ̊ ɰ̊ ɯ̊ ɱ̊ ɲ̊ ɳ̊ ɴ̊ ɵ̊ œ̊ ɪ̊ ʏ̊ ɾ̊ ɹ̊ ʂ̊ ʃ̊ ʊ̊ ʌ̊ ʎ̊ ʏ̊ ʐ̊ ʑ̊ ʒ̊ ʝ̊ ɮ̊

ą ƀ ç đ ę f ɡ ɨ̨ ɟ̨ k̨ l̨ m̨ n̨ ǫ p̨ q̨ ɾ̨ s̨ t̨ ů v̨ w̨ x̨ y̨ ƶ æ̨ œ̨ ŋ̨ ɛ̨ ɑ̨ ɔ̨ ɓ̨ ɖ̨ ƒ̨ ə̨ ə̨ ɛ̨ ɜ̨ ɛ̨ ɟ̨ ɡ̨ ɡ̨ ɡ̨ ɣ̨ ɣ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɰ̨ ɯ̨ ɱ̨ ɲ̨ ɳ̨ ɴ̨ ɵ̨ œ̨ ɪ̨ ʏ̨ ɾ̨ ɹ̨ ʂ̨ ʃ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ʝ̨ ɮ̨

ą ƀ ç đ ę f ɡ ɨ̨ ɟ̨ k̨ l̨ m̨ n̨ ǫ p̨ q̨ ɾ̨ s̨ t̨ ů v̨ w̨ x̨ y̨ ƶ æ̨ œ̨ ŋ̨ ɛ̨ ɑ̨ ɔ̨ ɓ̨ ɖ̨ ƒ̨ ə̨ ə̨ ɛ̨ ɜ̨ ɛ̨ ɟ̨ ɡ̨ ɡ̨ ɡ̨ ɣ̨ ɣ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɰ̨ ɯ̨ ɱ̨ ɲ̨ ɳ̨ ɴ̨ ɵ̨ œ̨ ɪ̨ ʏ̨ ɾ̨ ɹ̨ ʂ̨ ʃ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ʝ̨ ɮ̨

ạ ƀ ç ḍ ẹ f ɡ̣ ị j ḳ ḷ ṃ ṇ ọ p ɾ̣ ṣ ṭ ụ v ẉ x̣ ỵ ẓ æ̣ œ̣ ŋ ɛ̣ ɑ̣ ɔ̣ ɓ ɖ ƒ ə̣ ə̣ ɛ̣ ɜ̣ ɛ̣ ɟ ɡ̣ ɡ̣ ɡ̣ ɣ̣ ɣ̣ ɥ̣ ɦ ɨ̣ ɨ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ œ̣ ɪ̣ ʏ̣ ɾ̣ ɹ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʎ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ

ạ ƀ ç ḍ ẹ f ɡ̣ ị j ḳ ḷ ṃ ṇ ọ p ɾ̣ ṣ ṭ ụ v ẉ x̣ ỵ ẓ æ̣ œ̣ ŋ ɛ̣ ɑ̣ ɔ̣ ɓ ɖ ƒ ə̣ ə̣ ɛ̣ ɜ̣ ɛ̣ ɟ ɡ̣ ɡ̣ ɡ̣ ɣ̣ ɣ̣ ɥ̣ ɦ ɨ̣ ɨ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ œ̣ ɪ̣ ʏ̣ ɾ̣ ɹ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʎ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ

ạ ƀ ç ḍ ẹ f ɡ̣ ị j ḳ ḷ ṃ ṇ ọ p ɾ̣ ṣ ṭ ụ v ẉ x̣ ỵ ẓ æ̣ œ̣ ŋ ɛ̣ ɑ̣ ɔ̣ ɓ ɖ ƒ ə̣ ə̣ ɛ̣ ɜ̣ ɛ̣ ɟ ɡ̣ ɡ̣ ɡ̣ ɣ̣ ɣ̣ ɥ̣ ɦ ɨ̣ ɨ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ œ̣ ɪ̣ ʏ̣ ɾ̣ ɹ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʎ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ

ạ ƀ ç ḍ ẹ f ɡ̣ ị j ḳ ḷ ṃ ṇ ọ p ɾ̣ ṣ ṭ ụ v ẉ x̣ ỵ ẓ æ̣ œ̣ ŋ ɛ̣ ɑ̣ ɔ̣ ɓ ɖ ƒ ə̣ ə̣ ɛ̣ ɜ̣ ɛ̣ ɟ ɡ̣ ɡ̣ ɡ̣ ɣ̣ ɣ̣ ɥ̣ ɦ ɨ̣ ɨ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ œ̣ ɪ̣ ʏ̣ ɾ̣ ɹ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʎ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ

ạ ƀ ç ḍ ẹ f ɡ̣ ị j ḳ ḷ ṃ ṇ ọ p ɾ̣ ṣ ṭ ụ v ẉ x̣ ỵ ẓ æ̣ œ̣ ŋ ɛ̣ ɑ̣ ɔ̣ ɓ ɖ ƒ ə̣ ə̣ ɛ̣ ɜ̣ ɛ̣ ɟ ɡ̣ ɡ̣ ɡ̣ ɣ̣ ɣ̣ ɥ̣ ɦ ɨ̣ ɨ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ œ̣ ɪ̣ ʏ̣ ɾ̣ ɹ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʎ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ

ạ ƀ ç ḍ ẹ f ɡ̣ ị j ḳ ḷ ṃ ṇ ọ p ɾ̣ ṣ ṭ ụ v ẉ x̣ ỵ ẓ æ̣ œ̣ ŋ ɛ̣ ɑ̣ ɔ̣ ɓ ɖ ƒ ə̣ ə̣ ɛ̣ ɜ̣ ɛ̣ ɟ ɡ̣ ɡ̣ ɡ̣ ɣ̣ ɣ̣ ɥ̣ ɦ ɨ̣ ɨ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ œ̣ ɪ̣ ʏ̣ ɾ̣ ɹ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʎ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ

Italic, Bases requiring anchors for common marks, Common combining marks

à ɓ ɕ ɖ è f ɡ ì ɟ ɦ ɨ̀ ɨ̀ ɲ ɔ̀ ɸ ɋ̀ ɾ̀ ʃ̀ ʈ ù ʋ̀ ʍ̀ ɤ̀ ʏ̀ ʒ̀ æ̀ œ̀ ŋ̀ ɐ̀ ɑ̀ ɒ̀ ɓ̀ ɖ̀ ɗ̀ ɘ̀ ə̀ ɛ̀ ɜ̀ ɞ̀ ɟ̀ ɠ̀ ɡ̀ ɢ̀ ɣ̀ ɤ̀ ɥ̀ ɦ̀ ɨ̀ ɨ̀ ɭ̀ ɮ̀ ɯ̀ ɱ̀ ɲ̀ ŋ̀ ɴ̀ ɵ̀ ɶ̀ ɹ̀ ɻ̀ ɽ̀ ʁ̀ ʃ̀ ʂ̀ ʊ̀ ʌ̀ ʎ̀ ʏ̀ ʐ̀ ʑ̀ ʒ̀ ɟ̀ ɭ̀

á ɓ ɕ ɗ é f ɡ í ɟ ɦ ɨ́ ɨ́ ɲ ɔ́ ɸ ɋ́ ɾ́ ʃ́ ʈ ú ʋ́ ʍ́ ɤ́ ʏ́ ʒ́ ǽ œ́ ŋ́ ɐ́ ɑ́ ɒ́ ɓ́ ɖ́ ɗ́ ɘ́ ə́ ɛ́ ɜ́ ɞ́ ɟ́ ɠ́ ɡ́ ɢ́ ɣ́ ɤ́ ɥ́ ɦ́ ɨ́ ɨ́ ɭ́ ɮ́ ɯ́ ɱ́ ɲ́ ŋ́ ɴ́ ɵ́ ɶ́ ɹ́ ɻ́ ɽ́ ʁ́ ʃ́ ʂ́ ʊ́ ʌ́ ʎ́ ʏ́ ʐ́ ʑ́ ʒ́ ɟ́ ɭ́

â ɓ ɕ ɗ ê f ɡ î ɟ ɦ ɨ̂ ɨ̂ ɲ ɔ̂ ɸ ɋ̂ ɾ̂ ʃ̂ ʈ û ʋ̂ ʍ̂ ɤ̂ ʏ̂ ʒ̂ æ̂ œ̂ ŋ̂ ɐ̂ ɑ̂ ɒ̂ ɓ̂ ɖ̂ ɗ̂ ɘ̂ ə̂ ɛ̂ ɜ̂ ɞ̂ ɟ̂ ɠ̂ ɡ̂ ɢ̂ ɣ̂ ɤ̂ ɥ̂ ɦ̂ ɨ̂ ɨ̂ ɭ̂ ɮ̂ ɯ̂ ɱ̂ ɲ̂ ŋ̂ ɴ̂ ɵ̂ ɶ̂ ɹ̂ ɻ̂ ɽ̂ ʁ̂ ʃ̂ ʂ̂ ʊ̂ ʌ̂ ʎ̂ ʏ̂ ʐ̂ ʑ̂ ʒ̂ ɟ̂ ɭ̂

ã ɓ ɕ ɗ ẽ f ɡ ĩ ɟ ɦ ɨ̃ ɨ̃ ɲ ɔ̃ ɸ ɋ̃ ɾ̃ ʃ̃ ʈ ũ ʋ̃ ʍ̃ ɤ̃ ʏ̃ ʒ̃ æ̃ œ̃ ŋ̃ ɐ̃ ɑ̃ ɒ̃ ɓ̃ ɖ̃ ɗ̃ ɘ̃ ə̃ ɛ̃ ɜ̃ ɞ̃ ɟ̃ ɠ̃ ɡ̃ ɢ̃ ɣ̃ ɤ̃ ɥ̃ ɦ̃ ɨ̃ ɨ̃ ɭ̃ ɮ̃ ɯ̃ ɱ̃ ɲ̃ ŋ̃ ɴ̃ ɵ̃ ɶ̃ ɹ̃ ɻ̃ ɽ̃ ʁ̃ ʃ̃ ʂ̃ ʊ̃ ʌ̃ ʎ̃ ʏ̃ ʐ̃ ʑ̃ ʒ̃ ɟ̃ ɭ̃

ā ɓ ɕ ɗ ē f ɡ ī ɟ ɦ ɨ̄ ɨ̄ ɲ ɔ̄ ɸ ɋ̄ ɾ̄ ʃ̄ ʈ ū ʋ̄ ʍ̄ ɤ̄ ʏ̄ ʒ̄ ǣ œ̄ ŋ̄ ɐ̄ ɑ̄ ɒ̄ ɓ̄ ɖ̄ ɗ̄ ɘ̄ ə̄ ɛ̄ ɜ̄ ɞ̄ ɟ̄ ɠ̄ ɡ̄ ɢ̄ ɣ̄ ɤ̄ ɥ̄ ɦ̄ ɨ̄ ɨ̄ ɭ̄ ɮ̄ ɯ̄ ɱ̄ ɲ̄ ŋ̄ ɴ̄ ɵ̄ ɶ̄ ɹ̄ ɻ̄ ɽ̄ ʁ̄ ʃ̄ ʂ̄ ʊ̄ ʌ̄ ʎ̄ ʏ̄ ʐ̄ ʑ̄ ʒ̄ ɟ̄ ɭ̄

å ɓ ɕ ɗ ɛ̊ f ɡ̊ ɪ̊ ɟ̊ ɦ̊ ɨ̊ ɨ̊ ɲ̊ ɔ̊ ɸ̊ ɋ̊ ɾ̊ ʃ̊ ʈ̊ ů ʋ̊ ʍ̊ ɤ̊ ʏ̊ ʒ̊ æ̊ œ̊ ŋ̊ ɐ̊ ɑ̊ ɒ̊ ɓ̊ ɖ̊ ɗ̊ ɘ̊ ə̊ ɛ̊ ɜ̊ ɞ̊ ɟ̊ ɠ̊ ɡ̊ ɢ̊ ɣ̊ ɤ̊ ɥ̊ ɦ̊ ɨ̊ ɨ̊ ɭ̊ ɮ̊ ɯ̊ ɱ̊ ɲ̊ ŋ̊ ɴ̊ ɵ̊ ɶ̊ ɹ̊ ɻ̊ ɽ̊ ʁ̊ ʃ̊ ʂ̊ ʊ̊ ʌ̊ ʎ̊ ʏ̊ ʐ̊ ʑ̊ ʒ̊ ɟ̊ ɭ̊

ą ɓ ç ɖ ę f ɡ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ ɓ ç ɖ ę f ɡ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

Italic patched, Bases requiring anchors for common marks, Common combining marks

à ƀ ɕ ɖ è f ɡ ì ɟ ɦ ɨ̀ ɱ ɲ ò ɸ ɋ ɾ ʂ ʈ ù ʋ ɯ̀ x ʏ ʑ æ œ ŋ ɛ̀ ɑ̀ ɒ̀ ɓ ɖ ɗ ɔ̀ ə̀ ɛ̀ ɜ̀ ɞ̀ ɟ ɡ̀ ɡ̀ ɔ̀ ɣ̀ ɤ̀ ʉ̀ ɦ ɬ ɬ ɭ ɮ ɯ̀ ɱ ɲ ɳ ɴ ɵ̀ œ̀ ɹ̀ ɻ̀ ɽ̀ ʁ̀ ʂ ʃ ʊ̀ ʌ̀ ʎ̀ ʏ̀ ʐ̀ ʑ̀ ʒ̀ ɟ̀ ɭ

á ƃ ć ɗ é f ɡ í ɟ ɦ ɨ́ ɱ ɲ ó ɸ ɋ ɾ ś ʈ ú ʋ ɯ́ x ʏ ʑ ǽ œ ŋ ɛ́ ɑ́ ɒ́ ɓ ɖ ɗ ɔ́ ə́ ɛ́ ɜ́ ɞ́ ɟ ɡ́ ɡ́ ɔ́ ɣ́ ɤ́ ʉ́ ɦ ɬ ɬ ɭ ɮ ɯ́ ɱ ɲ ɳ ɴ ɵ́ œ́ ɹ́ ɻ́ ɽ́ ʁ́ ʂ ʃ ʊ́ ʌ́ ʎ́ ʏ́ ʐ́ ʑ́ ʒ́ ɟ́ ɭ

â ƀ ɕ ɖ ê f ɡ î ɟ ɦ ɨ̂ ɱ ɲ ô ɸ ɋ ɾ ŝ ʈ û ʋ ɯ̂ x ʏ ʑ ǽ œ ŋ ɛ̂ ɑ̂ ɒ̂ ɓ ɖ ɗ ɔ̂ ə̂ ɛ̂ ɜ̂ ɞ̂ ɟ ɡ̂ ɡ̂ ɔ̂ ɣ̂ ɤ̂ ʉ̂ ɦ ɬ ɬ ɭ ɮ ɯ̂ ɱ ɲ ɳ ɴ ɵ̂ œ̂ ɹ̂ ɻ̂ ɽ̂ ʁ̂ ʂ ʃ ʊ̂ ʌ̂ ʎ̂ ʏ̂ ʐ̂ ʑ̂ ʒ̂ ɟ̂ ɭ

ã ƀ ɕ ɖ ẽ f ɡ ĩ ɟ ɦ ɨ̃ ɱ ɲ õ ɸ ɋ ɾ s̃ ʈ ũ ʋ ɯ̃ x ʏ ʑ æ̃ œ̃ ŋ ɛ̃ ɑ̃ ɒ̃ ɓ ɖ ɗ ɔ̃ ə̃ ɛ̃ ɜ̃ ɞ̃ ɟ ɡ̃ ɡ̃ ɔ̃ ɣ̃ ɤ̃ ʉ̃ ɦ ɬ ɬ ɭ ɮ ɯ̃ ɱ ɲ ɳ ɴ ɵ̃ œ̃ ɹ̃ ɻ̃ ɽ̃ ʁ̃ ʂ ʃ ʊ̃ ʌ̃ ʎ̃ ʏ̃ ʐ̃ ʑ̃ ʒ̃ ɟ̃ ɭ

ā ƀ ɕ ɖ ē f ɡ ī ɟ ɦ ɨ̄ ɱ ɲ ō ɸ ɋ ɾ s̄ ʈ ū ʋ ɯ̄ x ʏ ʑ ǽ œ ŋ ɛ̄ ɑ̄ ɒ̄ ɓ ɖ ɗ ɔ̄ ə̄ ɛ̄ ɜ̄ ɞ̄ ɟ ɡ̄ ɡ̄ ɔ̄ ɣ̄ ɤ̄ ʉ̄ ɦ ɬ ɬ ɭ ɮ ɯ̄ ɱ ɲ ɳ ɴ ɵ̄ œ̄ ɹ̄ ɻ̄ ɽ̄ ʁ̄ ʂ ʃ ʊ̄ ʌ̄ ʎ̄ ʏ̄ ʐ̄ ʑ̄ ʒ̄ ɟ̄ ɭ

å ƀ ɕ ɖ ɛ̊ f ɡ̊ ɨ̊ ɟ ɦ ɨ̊ ɱ ɲ ɔ̊ ɸ ɋ ɾ s̊ ʈ ů ʋ ɯ̊ x ʏ ʑ æ̊ œ̊ ŋ ɛ̊ ɑ̊ ɒ̊ ɓ ɖ ɗ ɔ̊ ə̊ ɛ̊ ɜ̊ ɞ̊ ɟ ɡ̊ ɡ̊ ɔ̊ ɣ̊ ɤ̊ ʉ̊ ɦ ɬ ɬ ɭ ɮ ɯ̊ ɱ ɲ ɳ ɴ ɵ̊ œ̊ ɹ̊ ɻ̊ ɽ̊ ʁ̊ ʂ ʃ ʊ̊ ʌ̊ ʎ̊ ʏ̊ ʐ̊ ʑ̊ ʒ̊ ɟ̊ ɭ

ą ƀ ç ɖ ę f ɡ ɨ̨ ɟ ɦ ɨ̨ ɱ ɲ ǫ ɸ ɋ ɾ ʂ ʈ ų ʋ ɯ̨ x ʏ ʑ æ œ ŋ ɛ̨ ɑ̨ ɒ̨ ɓ ɖ ɗ ɔ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ ɡ̨ ɡ̨ ɔ̨ ɣ̨ ɤ̨ ʉ̨ ɦ ɬ ɬ ɭ ɮ ɯ̨ ɱ ɲ ɳ ɴ ɵ̨ œ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʂ ʃ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨

Semibold, Bases requiring anchors for common marks, Common combining marks

à ƀ ƈ ɖ è f ɠ ì ɟ ƙ ɭ ɱ ɲ ò ƥ ɋ ɾ ʃ ƭ ù ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɻ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

á ƀ ć d é f ɡ í ɟ ƙ ĺ ḿ ń ó ƥ ɋ ŕ ś ƭ ú ʋ ɯ ɤ ý ź ǽ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɽ

â ƀ ĉ d ê f ɡ î ɟ ƙ ĺ ḿ ń ô ƥ ɋ ŕ ś ƭ û ʋ ɯ ɤ ý ź æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ã ƀ ɕ d ẽ f ɡ ĩ ɟ ƙ ĺ ḿ ń õ ƥ ɋ ŕ ś ƭ ũ ʋ ɯ ɤ ý ź æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɭ

ā ƀ ɕ d ē f ɡ ī ɟ ƙ ĺ ḿ ń ō ƥ ɋ ŕ ś ƭ ū ʋ ɯ ɤ ý ź æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɽ

å ƀ ɕ d ɛ f ɡ ɨ ɟ ƙ l ɱ ɲ ɔ ƥ ɋ ŕ ś ƭ ů ʋ ɯ ɤ ý ź æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɭ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

Semibold patched, Bases requiring anchors for common marks, Common combining marks

à ƀ ƈ ɖ è f ɠ ì ɟ ƙ ǀ ɱ ɲ ò ƥ ɋ ɾ ʃ ʈ ù ʋ ɯ x ʏ ʒ æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɠ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɭ ɮ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ʟ

á ƃ ć d é f ɡ í ɟ ƙ ǁ ɱ ń ó ƥ ɋ ɾ ś ʈ ú ʋ ɯ x ʏ ź æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʑ ʒ ʝ ɮ

â ƃ ĉ d ê f ɡ î ɟ ƙ ǁ ɱ ɲ ô ƥ ɋ ɾ ŝ ʈ û ʋ ɯ x ʏ ẑ æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʑ ʒ ʝ ɮ

ã ƃ c̃ d ẽ f ɡ ĩ ɟ ƙ ǁ ɱ ñ õ ƥ ɋ ɾ s̃ ʈ ũ ʋ ɯ x ʏ z̃ æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ā ƃ c̄ d ē f ɡ ī ɟ ƙ ǁ ɱ n̄ ō ƥ ɋ ɾ s̄ ʈ ū ʋ ɯ x ʏ z̄ æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

å ƃ c̊ d ɛ̊ f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ɔ̊ ƥ ɋ ɾ ʃ ʈ ů ʋ ɯ x ʏ ʒ æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

Semibold italic, Bases requiring anchors for common marks, Common combining marks

Semibold italic patched, Bases requiring anchors for common marks, Common combining marks

à ƀ ȼ đ è f ǥ ì ɟ ƙ ǀ ɱ ǹ ò ƥ ɋ ɾ ſ ȶ ù ʋ ŵ x̀ ỳ ƶ æ œ ŋ ɐ̀ ɑ̀ ɒ̀ ɓ̀ ɖ̀ ƒ̀ ə̀ ɚ̀ ɛ̀ ɜ̀ ɘ̀ ɟ̀ ɡ̀ ɢ̀ ɣ̀ ɤ̀ ɥ̀ ɦ̀ ɨ̀ ɬ̀ ɭ̀ ɰ̀ ɯ̀ ɱ̀ ɲ̀ ɳ̀ ɴ̀ ɵ̀ ɶ̀ ɹ̀ ɻ̀ ɽ̀ ʂ̀ ʃ̀ ʊ̀ ʌ̀ ʏ̀ ʒ̀ ʑ̀ ʒ̀ ɟ̀ ɭ̀

á ƃ ć d́ é f́ ǵ í ǰ ḱ ĺ ḿ ń ó ƿ ɋ ŕ ś ť ú ʋ́ ẃ x́ ý ź ǽ œ́ ŋ́ ɐ́ ɑ́ ɒ́ ɓ́ ɖ́ ƒ́ ə́ ɚ́ ɛ́ ɜ́ ɘ́ ɟ́ ɡ́ ɢ́ ɣ́ ɤ́ ɥ́ ɦ́ ɨ́ ɬ́ ɭ́ ɰ́ ɯ́ ɱ́ ɲ́ ɳ́ ɴ́ ɵ́ ɶ́ ɹ́ ɻ́ ɽ́ ʂ́ ʃ́ ʊ́ ʌ́ ʏ́ ʒ́ ʑ́ ʒ́ ɟ́ ɭ́

â ƀ ĉ d̂ ê f ĝ î ĵ ƙ l̂ m̂ n̂ ô ƥ ɋ r̂ ŝ t̂ û ʋ ŵ x̂ ŷ ẑ æ œ ŋ ɐ̂ ɑ̂ ɒ̂ ɓ̂ ɖ̂ ƒ̂ ə̂ ɚ̂ ɛ̂ ɜ̂ ɘ̂ ɟ̂ ɡ̂ ɢ̂ ɣ̂ ɤ̂ ɥ̂ ɦ̂ ɨ̂ ɬ̂ ɭ̂ ɰ̂ ɯ̂ ɱ̂ ɲ̂ ɳ̂ ɴ̂ ɵ̂ ɶ̂ ɹ̂ ɻ̂ ɽ̂ ʂ̂ ʃ̂ ʊ̂ ʌ̂ ʏ̂ ʒ̂ ʑ̂ ʒ̂ ɟ̂ ɭ̂

ã ƀ c̃ d̃ ẽ f ɡ̃ ĩ ɟ̃ k̃ l̃ m̃ ñ õ ƥ ɋ r̃ s̃ t̃ ũ ʋ̃ w̃ x̃ ỹ z̃ æ œ ŋ ɐ̃ ɑ̃ ɒ̃ ɓ̃ ɖ̃ ƒ̃ ə̃ ɚ̃ ɛ̃ ɜ̃ ɘ̃ ɟ̃ ɡ̃ ɢ̃ ɣ̃ ɤ̃ ɥ̃ ɦ̃ ɨ̃ ɬ̃ ɭ̃ ɰ̃ ɯ̃ ɱ̃ ɲ̃ ɳ̃ ɴ̃ ɵ̃ ɶ̃ ɹ̃ ɻ̃ ɽ̃ ʂ̃ ʃ̃ ʊ̃ ʌ̃ ʏ̃ ʒ̃ ʑ̃ ʒ̃ ɟ̃ ɭ̃

ā ƀ c̄ d̄ ē f ɡ̄ ī ɟ̄ k̄ l̄ m̄ n̄ ō ƥ ɋ r̄ s̄ t̄ ū ʋ̄ w̄ x̄ ȳ z̄ æ œ ŋ ɐ̄ ɑ̄ ɒ̄ ɓ̄ ɖ̄ ƒ̄ ə̄ ɚ̄ ɛ̄ ɜ̄ ɘ̄ ɟ̄ ɡ̄ ɢ̄ ɣ̄ ɤ̄ ɥ̄ ɦ̄ ɨ̄ ɬ̄ ɭ̄ ɰ̄ ɯ̄ ɱ̄ ɲ̄ ɳ̄ ɴ̄ ɵ̄ ɶ̄ ɹ̄ ɻ̄ ɽ̄ ʂ̄ ʃ̄ ʊ̄ ʌ̄ ʏ̄ ʒ̄ ʑ̄ ʒ̄ ɟ̄ ɭ̄

ȧ ƀ ċ ḋ ė f ġ ȧ ɟ̇ ƙ l̇ ṁ ṅ ȯ ƥ ɋ ṙ ṡ ṫ u̇ ʋ̇ ẇ ẋ ẏ ż æ œ ŋ ɐ̇ ɑ̇ ɒ̇ ɓ̇ ɖ̇ ƒ̇ ə̇ ɚ̇ ɛ̇ ɜ̇ ɘ̇ ɟ̇ ɡ̇ ɢ̇ ɣ̇ ɤ̇ ɥ̇ ɦ̇ ɨ̇ ɬ̇ ɭ̇ ɰ̇ ɯ̇ ɱ̇ ɲ̇ ɳ̇ ɴ̇ ɵ̇ ɶ̇ ɹ̇ ɻ̇ ɽ̇ ʂ̇ ʃ̇ ʊ̇ ʌ̇ ʏ̇ ʒ̇ ʑ̇ ʒ̇ ɟ̇ ɭ̇

ạ ƀ ç ḍ ẹ f ɡ̣ ị ɟ̣ ḳ ḷ ṃ ṇ ọ ƥ ɋ ṛ ṣ ṭ ụ ʋ̣ ẉ ẋ ỵ ẓ æ œ ŋ ɐ̣ ɑ̣ ɒ̣ ɓ̣ ɖ̣ ƒ̣ ə̣ ɚ̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɬ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ ɶ̣ ɹ̣ ɻ̣ ɽ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʒ̣ ʑ̣ ʒ̣ ɟ̣ ɭ̣

ạ ƀ ç ḍ ẹ f ɡ̣ ị ɟ̣ ḳ ḷ ṃ ṇ ọ ƥ ɋ ṛ ṣ ṭ ụ ʋ̣ ẉ ẋ ỵ ẓ æ œ ŋ ɐ̣ ɑ̣ ɒ̣ ɓ̣ ɖ̣ ƒ̣ ə̣ ɚ̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɬ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ ɶ̣ ɹ̣ ɻ̣ ɽ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʒ̣ ʑ̣ ʒ̣ ɟ̣ ɭ̣

ạ ƀ ç ḍ ẹ f ɡ̣ ị ɟ̣ ḳ ḷ ṃ ṇ ọ ƥ ɋ ṛ ṣ ṭ ụ ʋ̣ ẉ ẋ ỵ ẓ æ œ ŋ ɐ̣ ɑ̣ ɒ̣ ɓ̣ ɖ̣ ƒ̣ ə̣ ɚ̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɬ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ ɶ̣ ɹ̣ ɻ̣ ɽ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʒ̣ ʑ̣ ʒ̣ ɟ̣ ɭ̣

ạ ƀ ç ḍ ẹ f ɡ̣ ị ɟ̣ ḳ ḷ ṃ ṇ ọ ƥ ɋ ṛ ṣ ṭ ụ ʋ̣ ẉ ẋ ỵ ẓ æ œ ŋ ɐ̣ ɑ̣ ɒ̣ ɓ̣ ɖ̣ ƒ̣ ə̣ ɚ̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɬ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ ɶ̣ ɹ̣ ɻ̣ ɽ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʒ̣ ʑ̣ ʒ̣ ɟ̣ ɭ̣

ạ ƀ ç ḍ ẹ f ɡ̣ ị ɟ̣ ḳ ḷ ṃ ṇ ọ ƥ ɋ ṛ ṣ ṭ ụ ʋ̣ ẉ ẋ ỵ ẓ æ œ ŋ ɐ̣ ɑ̣ ɒ̣ ɓ̣ ɖ̣ ƒ̣ ə̣ ɚ̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɬ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ ɶ̣ ɹ̣ ɻ̣ ɽ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʒ̣ ʑ̣ ʒ̣ ɟ̣ ɭ̣

ạ ƀ ç ḍ ẹ f ɡ̣ ị ɟ̣ ḳ ḷ ṃ ṇ ọ ƥ ɋ ṛ ṣ ṭ ụ ʋ̣ ẉ ẋ ỵ ẓ æ œ ŋ ɐ̣ ɑ̣ ɒ̣ ɓ̣ ɖ̣ ƒ̣ ə̣ ɚ̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɬ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ ɶ̣ ɹ̣ ɻ̣ ɽ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʒ̣ ʑ̣ ʒ̣ ɟ̣ ɭ̣

ạ ƀ ç ḍ ẹ f ɡ̣ ị ɟ̣ ḳ ḷ ṃ ṇ ọ ƥ ɋ ṛ ṣ ṭ ụ ʋ̣ ẉ ẋ ỵ ẓ æ œ ŋ ɐ̣ ɑ̣ ɒ̣ ɓ̣ ɖ̣ ƒ̣ ə̣ ɚ̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɬ̣ ɭ̣ ɰ̣ ɯ̣ ɱ̣ ɲ̣ ɳ̣ ɴ̣ ɵ̣ ɶ̣ ɹ̣ ɻ̣ ɽ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʒ̣ ʑ̣ ʒ̣ ɟ̣ ɭ̣

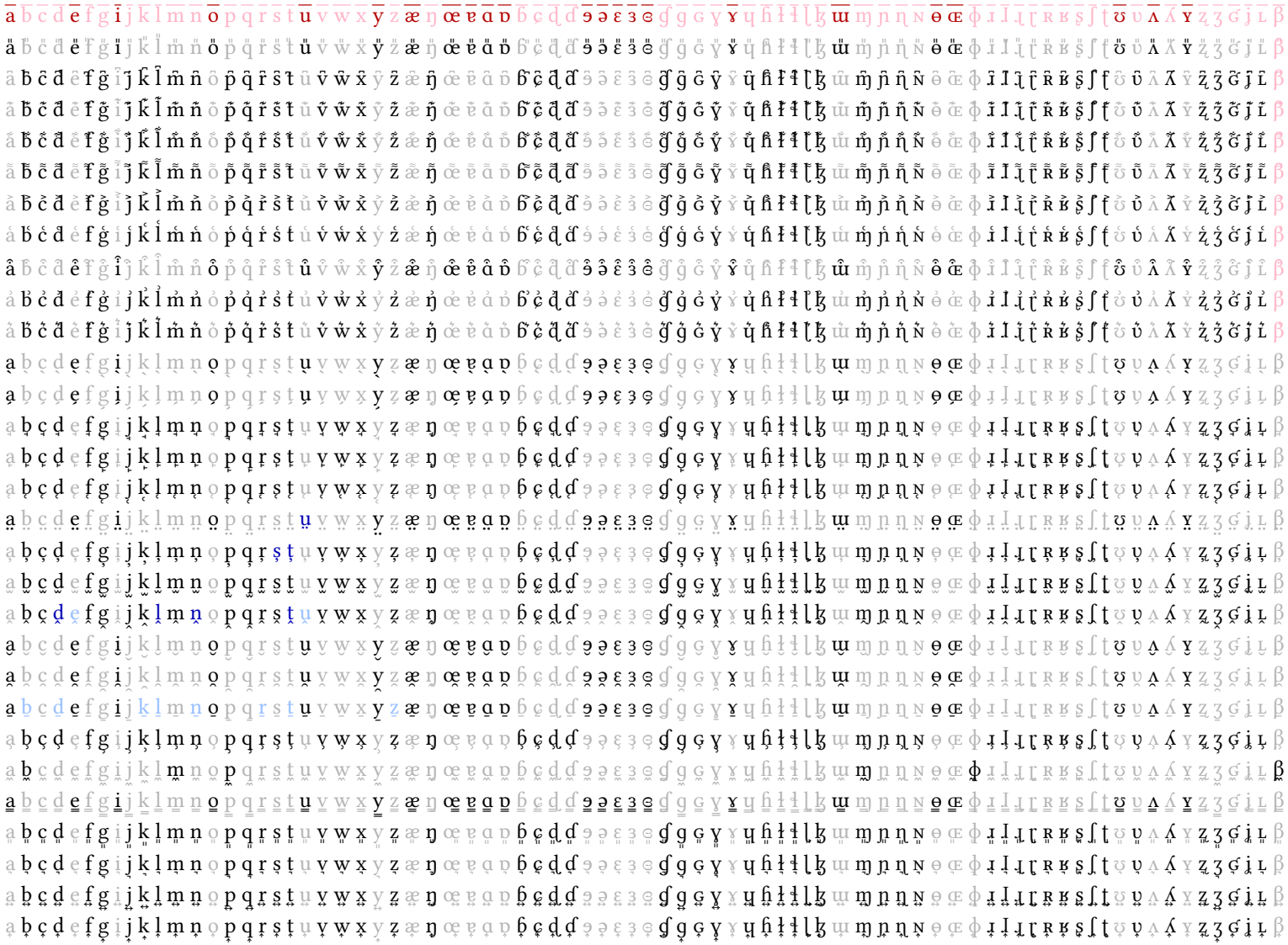
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɐ ɑ ɒ ɓ ɖ ƒ ə ɚ ɛ ɜ ɘ ɟ ɡ ɢ ɣ ɤ ɥ ɦ ɨ ɬ ɭ ɰ ɯ ɱ ɲ ɳ ɴ ɵ ɶ ɹ ɻ ɽ ʂ ʃ ʊ ʌ ʏ ʒ ʑ ʒ ɟ ɭ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɐ ɑ ɒ ɓ ɖ ƒ ə ɚ ɛ ɜ ɘ ɟ ɡ ɢ ɣ ɤ ɥ ɦ ɨ ɬ ɭ ɰ ɯ ɱ ɲ ɳ ɴ ɵ ɶ ɹ ɻ ɽ ʂ ʃ ʊ ʌ ʏ ʒ ʑ ʒ ɟ ɭ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɐ ɑ ɒ ɓ ɖ ƒ ə ɚ ɛ ɜ ɘ ɟ ɡ ɢ ɣ ɤ ɥ ɦ ɨ ɬ ɭ ɰ ɯ ɱ ɲ ɳ ɴ ɵ ɶ ɹ ɻ ɽ ʂ ʃ ʊ ʌ ʏ ʒ ʑ ʒ ɟ ɭ

Regular, Bases requiring anchors for rare marks, Rare combining marks

Regular patched, Bases requiring anchors for rare marks, Rare combining marks



Italic, Bases requiring anchors for rare marks, Rare combining marks

Italic patched, Bases requiring anchors for rare marks, Rare combining marks

Semibold, Bases requiring anchors for rare marks, Rare combining marks

Semibold patched, Bases requiring anchors for rare marks, Rare combining marks

Semibold italic, Bases requiring anchors for rare marks, Rare combining marks

Semibold italic patched, Bases requiring anchors for rare marks, Rare combining marks

Patch superscript consonant anchors

Libertinus serif regular

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ᵐ ᵐ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ᵐ ᵐ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ᵐ ᵐ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

Libertinus serif regular, patched

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ᵐ ᵐ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ᵐ ᵐ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ᵐ ᵐ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

Gentium Plus

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ᵐ ᵐ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ᵐ ᵐ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

ᵐ ᵐ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵘ ᵛ ᵊ ᵝ ᵞ ᵟ

A ᵃ B ᵇ C ᶜ D ᵈ E ᵉ F ᶠ G ᵍ H ʰ I ᵢ J ʲ K ᵏ L ˡ M ᵐ N ᵒ O ᵒ P ᵖ Q ᵠ R ʳ S ˢ T ᵗ U ᵘ V ᵛ W ʷ X ˣ Y ʸ Z ᶻ a ᵃ b ᵇ c ᶜ d ᵈ e ᵉ f ᶠ g ᵍ h ʰ i ᵢ j ʲ k ᵏ l ˡ m ᵐ n ᵒ o ᵒ p ᵖ q ᵠ r ʳ s ˢ t ᵗ u ᵘ v ᵛ w ʷ x ˣ y ʸ z ᶻ Æ ᵃᵉ æ ᵃᵉ Œ ᵒᵉ œ ᵒᵉ Ø ᵒᵉ ø ᵒᵉ Ð ᵈʰ ð ᵈʰ Þ ᵗʰ þ ᵗʰ ß ˢˢ S ˢˢ Đ ᵈʰ đ ᵈʰ Ħ ʰʰ ħ ʰʰ ı ᵢ ȷ ʲ Ł ˡˡ ƛ ˡˡ Б ᵇᵇ

À B̀ Ć D̀ È F̀ Ǵ H̀ Ì J̀ Ḱ L̀ M̀ Ń Ò P̀ Q́ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Ż à b̀ ć d̀ è f̀ ǵ ḥ ì j̀ ḱ l̀ m̀ ń ò p̀ q́ r̀ s̀ t̀ ù v̀ ẁ x̀ ỳ ż Æ̀ æ̀ Œ̀ œ̀ Ǿ ǿ Ð̀ ð̀ Þ̀ þ̀ ß̀ ſ́ Đ̀ đ̀ Ħ̀ ħ̀ ì j̀ Ł̀ ł̀ Ƀ̀ ƀ̀